

# Closure Forces Structure:

## The Standard Model from Rosen Closure on Ternary Causal Hypergraphs

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*“If you know your way broadly, you will see it in all things.”* — Musashi

### Abstract

Rosen’s closure to efficient causation—the requirement that every functional component of a system is produced by the system itself—is formalized as a fixed-point equation in a Cartesian Closed Category (CCC). We realize this fixed point in the category TCHyp of ternary causal hypergraphs with double-pushout (DPO) rewriting and show that a chain of algebraic obstructions eliminates every decoration structure except one: the gauge group  $[\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y] \times \mathrm{SU}(3)_{\mathrm{gen}}$  and the Born-rule probability measure, with zero free structural parameters. (The 224 vacuum moduli—mass ratios, mixing angles, CP phases—are free within the spectral action minimum.)

A truncation functor  $T_2$  from any CCC with a reflexive object to TCHyp shows that TCHyp is not an independent modeling assumption but the canonical output of Rosen closure: the same obstruction chain holds in any adhesive category with DPO dynamics satisfying a mild representability hypothesis.

The gauge-equivalence quotient  $Q_{24} = M(G_0)/\text{gauge}$  collapses the infinite multi-way graph to a finite 24-vertex autopoietic fixed point. Its C-closure  $Q_{48} = Q_{24} \cup C(Q_{24})$  carries a KO-dimension 6 spectral triple whose Connes–Chamseddine–Marcolli (CCM) classification forces the algebra  $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ , the lepton sector, and anomaly cancellation—resolving the single bridge principle of the obstruction chain without external input.

The spacetime-scale quotient  $Q_{102} = Q_{51} \cup C(Q_{51})$  unifies gauge and gravitational degrees of freedom in a single Dirac operator whose  $\gamma$ -decomposition  $D = D_+ + D_-$  derives the Connes product structure, a discrete Coleman–Mandula theorem, and an explicit gauge-covariant gravity-gauge coupling operator—the cross term  $C = D_+ D_- + D_- D_+$ —that enters the spectral action at fourth order.  $Q_{102}$  is a self-reproducing fixed

point under iterated DPO composition, making it the minimal self-sustaining unit of the theory.

The formal specification contains 151 scorecard entries: 130 proved unconditional and 21 computationally verified. Nine entries have categorical machine proofs via CatLab.jl. All proofs are deductive from two principles: organisational closure and a representability hypothesis.

# 1 Introduction

## 1.1 The Result and Its Scope

Aristotle distinguished four modes of causation: material, formal, efficient, and final. Modern science retained efficient causation and discarded the rest. Rosen [21] observed that self-maintaining systems require a stronger condition: every efficient cause within the system must itself be produced by the system. This is *closure to efficient causation*. Hofmeyr [13, 14] formalized it in the biochemical setting as fabrication-assembly networks. Here we show that the same principle, formalized as a fixed-point equation in a Cartesian Closed Category, constrains the space of possible self-maintaining systems at any scale—and, when realized in a specific category of causal hypergraphs, forces the Standard Model gauge group.

The argument requires two inputs:

**Principle 1 (Closure).** The system is closed to efficient causation: every functional component is produced by the system itself.

**Principle 2 (Realization).** The closure is realized in a specific category of hypergraph rewriting systems with double-pushout dynamics.

Principle 2 was originally a substantive modelling choice. The universality theorem (Section 9, proved via the truncation functor  $T_2$ ) shows that it is in fact a consequence of Principle 1: TCHyp is the canonical output of Rosen closure in any CCC with a reflexive object, and the same obstruction chain holds in any adhesive category with DPO dynamics satisfying a representability hypothesis strictly weaker than Principle 2 itself. Principle 2 is retained for expository clarity but is no longer an independent assumption.

The following structures are forced:

- (a) Exactly three functional roles (metabolism, repair, replication).
- (b) Hyperedge arity  $\tau = 3$ .
- (c) The impossibility of discrete decoration labels (proved for all finite targets with any ternary composition rule).
- (d) The gauge group  $[\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y] \times \mathrm{SU}(3)_{\mathrm{gen}}$ .

- (e) The Born-rule probability measure  $\mu = |\det[\tilde{\psi}]|^2$ .
- (f) Charge conjugation  $C$  with forced  $C_w = i\sigma_2$ .
- (g) Maximal structural parity violation.
- (h) Generic CP violation sourced from the complex decoration field.
- (i) A finite gauge-equivalence quotient  $Q_{24}$  (autopoietic).
- (j) A KO-dimension 6 spectral triple on  $Q_{48}$  forcing leptons and anomaly cancellation.
- (k) A unified gauge-gravity Dirac operator on  $Q_{102}$  deriving the Connes product structure,  $d_s \sim 4$  spacetime, discrete gravity, and gauge-gravity coupling.
- (l) Self-reproduction:  $Q_{102}$  is a fixed point under its own rewriting dynamics.

## 1.2 Foundational Logic and Layer Structure

The theory operates in three distinct logical layers. These are not interchangeable; conflating them produces category errors.

**Possibilistic Layer (Layers 1–7 of the formal specification).** A modal logic of the constraint surface with three operators: FORCED (necessarily true for all Rosen-closed systems in TCHyp), FORBIDDEN (structurally impossible under Rosen closure), and COMPATIBLE (structurally allowed but not uniquely determined). This layer is static. Classical category theory is sufficient. The entire gauge group derivation lives here.

**Probabilistic Layer (Layer 7+).** Assigns a measure-theoretic structure to compatible configurations: the Born rule  $\mu(B) = |\det M_B|^2$ . Requires the possibilistic structure as input but adds Gleason’s theorem. The Born rule, spectator singlet-projection, and measurement identification live here.

**Bridge Layer (now unconditional).** Originally conditional on anomaly cancellation (Bridge Principle B1). The NCG route (Section 11) derives B1 from the spectral triple on  $Q_{48}$ , making this layer fully unconditional. Contains the lepton sector and hypercharge determination.

## 1.3 Architecture of the Proof

The theorem chain has seven links. Each is a forcing result: given the output of the previous link, the next structure is the unique possibility compatible with closure.

|                                                              |                        |
|--------------------------------------------------------------|------------------------|
| Axiom R (CCC fixed point)                                    |                        |
| $\Rightarrow$ Three roles $\{F, A, S\}$                      | [Theorem 3.1, Sec 3]   |
| $\Rightarrow$ Arity $\tau = 3$                               | [Theorem 4.1, Sec 4]   |
| $\Rightarrow$ Discrete decorations impossible                | [Theorem 5.2, Sec 5]   |
| $\Rightarrow$ Composition on $F^3$ , cross-product unique    | [Hurwitz, Sec 5.5]     |
| $\Rightarrow k = 7$ eliminated                               | [Lemma 5.4, Sec 5.5]   |
| $\Rightarrow F = \mathbb{C}$ (self-referential isotropy)     | [Theorem 5.5, Sec 5.6] |
| $\Rightarrow$ Weak $= \mathbb{C}^2$ , gauge $= \text{SU}(2)$ | [Theorem 5.7, Sec 5.7] |
| $\Rightarrow$ Weak composition unique (Schur)                | [Theorem 5.8, Sec 5.7] |
| $\Rightarrow$ Gauge group derived                            | [Theorem 6.1, Sec 6]   |
| $\Rightarrow$ Born rule derived                              | [Theorem 7.1, Sec 7]   |
| $\Rightarrow$ Discrete symmetries $C, P, CP$ derived         | [Sec 7.6]              |
| — B1 (derived via NCG, Sec 11) —                             |                        |
| $\Rightarrow$ Lepton sector + hypercharges                   | [Sec 7.5]              |

Universality (Theorem 9.2, Sec 9):  $T_2$  functor produces  $G_0 + G'_2$ ; representability hypothesis sufficient; entire chain holds universally; Axiom  $T$  (Principle 2) eliminated.

NCG Resolution (Sec 11):  $Q_{48}$  spectral triple (KO-dim 6)  $\Rightarrow$  CCM classification  $\Rightarrow$  leptons forced  $\Rightarrow$  B1 derived (no external input).

Gauge-Spacetime Unification (Sec 12):  $Q_{102}$  spectral triple  $\Rightarrow$   $\gamma$ -orthogonality  $\Rightarrow$  product structure  $\Rightarrow$  discrete Coleman–Mandula  $\Rightarrow$  self-reproducing fixed point.

## 1.4 Relation to Prior Work

Rosen [21] introduced closure to efficient causation as the defining property of living systems. Mossio, Longo, and Stewart [18] showed that the  $(M, R)$  closure condition lives naturally in a CCC via Scott’s  $D_\infty$  construction. Hofmeyr [14] gave a biochemically concrete realization as the  $(F, A)$ -Sys fabrication-assembly architecture. Gorard [11, 12] developed the DPO rewriting framework for causal hypergraphs.

The present paper shows that combining the Mossio–Longo CCC formulation with Gorard’s rewriting framework yields, by pure obstruction theory, the gauge structure of the Standard Model. No physics is assumed. The gauge group appears as the automorphism group of the unique decoration functor compatible with closure. The universality theorem shows this result is not specific to TCHyp.

The Connes–Chamseddine–Marcolli programme [8, 9, 6, 7] derives the SM Lagrangian from a spectral triple on  $M^4 \times F$ , with the finite geometry  $F$  chosen to reproduce the SM. Our contribution is to derive  $F$  itself—the finite spectral triple arises from closure dynamics on ternary hypergraphs, not from a choice of algebra.

The gauge non-covariance of Einstein’s curvature projection—that  $P_E$  does not commute with gauge transformations [23, 15]—is a central obstruction to geometric unification. The spectral action on the closure-derived finite geometry automatically generates a gauge-covariant gravity-gauge coupling operator  $C = D_+ D_- + D_- D_+$ , resolving this obstruction

without being designed to do so. The operator  $C$  has the schematic form proposed independently by Weinstein [24] under the name “Shiab operator,” though our construction is explicit and fully computed whereas that proposal remained incomplete.

## 2 Preliminaries

### 2.1 Cartesian Closed Categories

A Cartesian Closed Category (CCC)  $\mathcal{C}$  is a category with finite products and, for every pair of objects  $A, B$ , an exponential object  $B^A$  together with an evaluation morphism  $\text{eval} : B^A \times A \rightarrow B$  satisfying the universal property: for every morphism  $g : C \times A \rightarrow B$ , there exists a unique morphism  $\lambda(g) : C \rightarrow B^A$  such that  $\text{eval} \circ (\lambda(g) \times \text{id}_A) = g$ .

The exponential  $B^A$  internalizes the hom-set  $\text{Hom}(A, B)$  as an object of  $\mathcal{C}$ . This permits self-reference: morphisms can be treated as objects, and morphisms-on-morphisms as objects of objects.

### 2.2 Reflexive Objects and the $D_\infty$ Construction

A reflexive object in a CCC  $\mathcal{C}$  is an object  $D$  equipped with a retraction pair:

$$i : D^D \rightarrow D, \quad j : D \rightarrow D^D, \quad j \circ i = \text{id}_{D^D}.$$

Scott [22] constructs such an object as the inverse limit  $D_\infty = \varprojlim D_n$  of the tower

$$D_0 \leftarrow D_1 \leftarrow D_2 \leftarrow \dots$$

where  $D_{n+1} = D_n^{D_n}$ . The limit satisfies  $D_\infty \cong D_\infty^{D_\infty}$ : it is isomorphic to its own function space.

### 2.3 Rosen’s (M,R) System

Rosen [21] defines a system closed to efficient causation as a triple of morphisms forming a dependency loop:

$$\begin{array}{ll} f : A \rightarrow B & \text{(metabolism)} \\ \Phi : B \rightarrow \text{Hom}(A, B) & \text{(repair: produces } f \text{ from outputs)} \\ \beta : \text{Hom}(A, B) \rightarrow \text{Hom}(B, \text{Hom}(A, B)) & \text{(replication: produces } \Phi \text{ from } f) \end{array}$$

with  $\beta$  itself entailed by the system.

Mossio, Longo, and Stewart [18] observed that this self-referential structure requires a reflexive object in a CCC:

$$D \cong D^D \Rightarrow f \in D, \quad \Phi \in D^D \cong D, \quad \beta \in D^{D^D} \cong D.$$

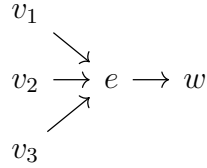
The closure condition is the fixed-point equation: the system’s components must be members of an object that is isomorphic to its own function space.

FIGURE 1: Rosen  $(M, R)$  closure diagram showing  $f, \Phi, \beta$  dependency loop with CCC exponential stru

Figure 1: The Rosen  $(M, R)$  closure diagram.

## 2.4 The Category TCHyp

A ternary causal hypergraph is a tuple  $(V, E, s, t)$  where  $V$  is a finite set of vertices,  $E$  is a finite set of hyperedges,  $s : E \rightarrow V^3$  assigns an ordered source triple, and  $t : E \rightarrow V$  assigns a target vertex.



$$s(e) = (v_1, v_2, v_3), \quad t(e) = w.$$

Position 1 =  $F$ -role, 2 =  $A$ -role, 3 =  $S$ -role.

A morphism in TCHyp is a pair of functions  $(f_V : V \rightarrow V', f_E : E \rightarrow E')$  preserving source and target maps.

## 2.5 Adhesive Categories and Double-Pushout Rewriting

TCHyp is adhesive: pushouts along monomorphisms satisfy the Van Kampen property [16]. This ensures that double-pushout (DPO) rewriting is well-defined with unique pushout complements.

$$\begin{array}{ccccc} L & \longleftarrow & K & \longrightarrow & R \\ \uparrow & & \uparrow & & \uparrow \\ G & \longleftarrow & D & \longrightarrow & H \end{array}$$

$L$  = left-hand side (pattern to match),  $K$  = interface (preserved structure),  $R$  = right-hand side (replacement),  $G$  = host graph,  $H$  = result graph,  $D$  = context (pushout complement).

**Proposition 2.1** (TCHyp adhesive). *TCHyp has pushouts along monomorphisms satisfying the Van Kampen property. Pushout complements exist and are unique for all DPO matches on  $G_0$ .*<sup>S25</sup>

**Proposition 2.2** (Church–Rosser). *Parallel-independent DPO rewrites in TCHyp produce isomorphic results.*<sup>S26</sup>

## 2.6 Decoration Functors

A decoration functor  $\text{Dec} : \text{TCHyp} \rightarrow \text{Vect}_{\mathbb{C}}$  assigns to each vertex  $v$  a vector  $\psi(v)$  in some target space  $F$ , and to each hyperedge  $e$  a composition rule  $\mu : F \times F \times F \rightarrow F$ . Closure requires the composition to be self-consistent: the target vertex carries the same type of object as the source vertices.

The gauge group is  $\text{Aut}(\text{Dec})$ —the group of natural transformations of the decoration functor that commute with composition. The central result of this paper is that  $\text{Aut}(\text{Dec})$  is uniquely determined by closure.

## 3 The Three-Role Theorem

**Theorem 3.1** (Three Rosen roles). *In any CCC with a reflexive object  $D \cong D^D$ , the minimal Rosen-closed system has exactly three functional roles: metabolism ( $f$ ), repair ( $\Phi$ ), and replication ( $\beta$ ).<sup>S1</sup>*

*Proof sketch.* The exponential tower  $D^D \cong D$  forces three levels: Level 0 ( $f \in D$ ), Level 1 ( $\Phi \in D^D \cong D$ ), Level 2 ( $\beta \in D^{D^D} \cong D$ ). A fourth morphism at any level is either (a) absorbed into the existing system via the universal property of the exponential, or (b) constitutes a separate three-role system. The uniqueness of the exponential transpose  $\lambda$  ensures exactly one self-entailing morphism at Level 2 given  $(f, \Phi)$ .  $\square$

The three roles correspond to the three source positions of a ternary hyperedge: position 1 ( $F$ -role, metabolism), position 2 ( $A$ -role, repair), position 3 ( $S$ -role, replication/spectator).

## 4 Ternary Arity

**Theorem 4.1** (Arity  $\tau = 3$ ). *The minimal arity compatible with Rosen closure and the Born-rule probability measure is  $\tau = 3$ .<sup>S2</sup>*

Three independent proofs:

*Proof 1 (Role independence, Lemma 4.1).* For  $n < 3$ , at least two Rosen roles must share a source position. The shared role cannot independently vanish—the multiplicative persistence gate requires independent zeros. Binary hypergraphs produce only 2 edges per rewrite; there is no spectator position.<sup>S27</sup>  $\square$

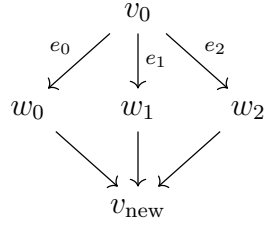
*Proof 2 (Gleason obstruction, Lemma 4.2).* Gleason’s theorem requires  $\dim \geq 3$  for a unique probability measure on Hilbert space. With  $\tau = 2$ , the decoration space is  $\mathbb{C}^2$ , where Gleason’s theorem fails (Kochen–Specker counterexamples exist). With  $\tau = 3$ , the decoration space is  $\mathbb{C}^3$ , where Gleason’s theorem guarantees the Born rule is the unique frame function.<sup>S28</sup>  $\square$

*Proof 3 (Sufficiency, Lemma 4.3).* Ternary DPO produces exactly 3 daughter edges with a spectator position, the minimum needed for self-referential closure.<sup>S29</sup>  $\square$

## 5 The Impossibility of Discrete Decorations

### 5.1 The Categorical Framework

The DPO rewrite rule for a single ternary production has the structure:



The self-referential graph  $G_0$  is the minimal structure that is both the pattern and the host: the system rewrites itself.

**Definition 5.1** (Self-referential graphs  $G_0$  and  $G'_2$ ).  $G_0$  is the minimal Rosen-closed ternary hypergraph.  $G'_2$  is a companion graph sharing the daughter source triple  $(w, v_0, v_1)$  with  $G_0$  but targeting a different vertex. Together they form a graph pair whose self-referential constraints force identical entries in certain positions.

### 5.2 Specific Obstructions

Each discrete group is eliminated by a specific obstruction:

- (a)  $\mathbb{Z}_3$  colour: the composition  $\mu(a, b, c)$  must be equivariant under the colour group, but the self-referential constraint forces  $a = c$  on the source triple, collapsing  $\mathbb{Z}_3$  to the trivial group.<sup>S3</sup>
- (b)  $\mathbb{Z}_2$  weak: the  $SU(2)$  holonomy restriction to a single vertex via the stabilizer chain forces the  $\mathbb{Z}_2$  subgroup to act trivially.<sup>S4</sup>
- (c)  $\mathbb{Z}_G$  generation ( $G = 1, \dots, 8$ ): exhaustive verification for all composition rules on  $\mathbb{Z}_G$  shows no Rosen-closed solution.<sup>S6</sup>
- (d) Spinor ( $\mathbb{Z}_2$  cover): the double cover of  $SO(3)$  cannot be realized as a decoration of a ternary hypergraph with DPO dynamics.<sup>S31</sup>



### 5.3 The General Discrete Obstruction

**Theorem 5.2** (General discrete obstruction). *For any finite target set  $A$  with any ternary composition rule  $\mu : A^3 \rightarrow A$ , there exists no Rosen-closed decoration of  $G_0$  under DPO rewriting.*<sup>S6a</sup>

*Proof (Theorem 5.9b).* The graph pair  $(G_0, G'_2)$  share the daughter source triple but target different vertices. Their Rosen closure constraints, intersected, force  $a = c$  on every admissible source triple  $(a, b, c)$ . But independent vanishing of coordinate 0 requires  $a \neq c$  for at least one triple. This is a contradiction for *any* finite  $A$ , *any*  $\mu$ , *any* branching degree  $m \geq 2$ . The argument is purely combinatorial: it uses only the graph structure, not the composition rule.  $\square$

This is the foundational result. It eliminates all discrete decoration targets and forces continuous structures downstream. Proved for all finite  $A$  by graph-pair identity (v14). The earlier exhaustive verifications for  $|A| \leq 2$  and  $\mathbb{Z}_G$  ( $G = 1-8$ ) are independent confirmations. Lemma 7.0a extends this: the graph-pair identity is cardinality-independent, killing countably infinite discrete targets as well.

### 5.4 The Unified Pattern

The obstruction pattern is uniform: every discrete decoration attempt fails at the same structural point—the self-referential graph  $G_0$  forces equations that overdetermine the system. The composition rule must satisfy closure constraints at every vertex, but these constraints form an inconsistent system over any discrete target.

### 5.5 The Hurwitz Constraint and $k = 7$ Elimination

Since discrete targets are impossible, the decoration target must be a continuous space. The composition rule  $\mu : F^3 \rightarrow F$  must be bilinear (from the cross-product structure forced by closure) and norm-preserving (from the unit-norm constraint on decorations).

**Lemma 5.3** (Hurwitz). *The only normed division algebras over  $\mathbb{R}$  are  $\mathbb{R}, \mathbb{C}, \mathbb{H}$ , and  $\mathbb{O}$ , with dimensions  $k \in \{1, 2, 4, 8\}$ . The bilinear cross-product dimension is  $k - 1 \in \{0, 1, 3, 7\}$ .*<sup>S139</sup>

The  $k = 0$  case is trivial (scalar, no structure). The remaining candidates are  $k = 1$  ( $\mathbb{R}$ , cross-product trivial),  $k = 3$  ( $\mathbb{C}^3$ , cross-product), and  $k = 7$  ( $\mathbb{O}^7$ , octonionic cross-product).

**Lemma 5.4** ( $k = 7$  elimination). *The octonionic case  $k = 7$  is eliminated by three independent incompatibilities:*

- (a) *Reducible stabilizer.*  $\text{Stab}_{G_2}(\hat{n}) \cong \text{SU}(3)$  acts on  $\mathbb{C}^6 = \mathbf{3} \oplus \bar{\mathbf{3}}$  (reducible), breaking the Schur obstruction needed for the weak sector.

(b) *Vacuous closure phase.*  $G_2 \subset \mathrm{SU}(7)$ —the determinant constraint is automatically satisfied, providing no structure.

(c) *No abelian factor.*  $G_2$  is simple—no  $\mathrm{U}(1)$  subgroup exists for hypercharge.

*Each incompatibility is independently fatal.*<sup>S139</sup>

## 5.6 Self-Referential Isotropy: $F = \mathbb{C}$

**Theorem 5.5** ( $F = \mathbb{C}$ ). *The decoration field is the complex numbers, not the reals.*<sup>S3a</sup>

*Proof.* The BAC-CAB identity on the self-referential edge  $e_3$  of  $G_0$  gives:  $a \times (b \times a) = b|a|^2 - a(a \cdot b)$ . Over  $\mathbb{R}$ , this forces isotropic unit vectors ( $|a|^2 = 1$  and  $a \cdot a = 0$  simultaneously), which do not exist. Over  $\mathbb{C}$ , the inner product is Hermitian:  $a \cdot a = |a|^2 \neq 0$  but  $a^T a$  can vanish (complex isotropy). The solution space over  $\mathbb{C}$  has dimension 11 with 0 physical degrees of freedom (all gauged).  $\square$

This result has a deep consequence: the complex field that makes quantum mechanics possible is the *same* structure that forces CP violation (Proposition 7.11). The mechanism that creates superposition simultaneously creates matter-antimatter asymmetry.

*Remark 5.6.* The discrete obstruction (Theorem 5.2) eliminates all finite and countable decoration targets. The step from “continuous target required” to “compact connected manifold” invokes Lemma 7.0b, which remains open (see Section 14.3). Without it, the argument shows that  $\mathbb{C}$  works and that no discrete alternative exists, but does not prove that  $\mathbb{C}$  is the *unique* continuous solution.

## 5.7 Colour Stabilizer: $\text{Weak} = \mathbb{C}^2$ with $\mathrm{SU}(2)$ Gauge

**Theorem 5.7** ( $k_{\text{weak}} = 2$ ). *The weak decoration space is  $\mathbb{C}^2$  with gauge group  $\mathrm{SU}(2)$ .*<sup>S4a</sup>

*Proof.* The colour stabilizer chain:  $\mathrm{Stab}_{\mathrm{SU}(3)}(\hat{n}) \cong \mathrm{SU}(2)$ , where  $\hat{n}$  is the axis of the cross product. The stabilizer acts on the perpendicular plane  $\hat{n}^\perp \cong \mathbb{C}^2$  in the fundamental representation. This is the unique faithful 2-dimensional representation of  $\mathrm{SU}(2)$ .  $\square$

**Theorem 5.8** (Weak composition unique). *The weak sector composition is uniquely determined by the colour sector.*<sup>S4b</sup>

*Proof.* The Schur obstruction:  $\dim \mathrm{Hom}_{\mathrm{SU}(2)}(\mathbf{2} \otimes \mathbf{2}, \mathbf{2}) = 0$ . There is no bilinear  $\mathrm{SU}(2)$ -equivariant map  $\mathbb{C}^2 \times \mathbb{C}^2 \rightarrow \mathbb{C}^2$ . The only equivariant generator is the determinant (antisymmetric,  $\dim \mathrm{Hom} = 1$ ). Therefore the weak composition is uniquely fixed by equivariance—the colour cross-product axis determines everything.  $\square$

## 6 The Gauge Group

### 6.1 Derivation

**Theorem 6.1** (Gauge group). *The automorphism group of the decoration functor compatible with Rosen closure on TCHyp is:*

$$\text{Aut}(\text{Dec}) = [\text{SU}(3)_{\text{col}} \times \text{SU}(2)_{\text{weak}} \times \text{U}(1)_Y] \times \text{SU}(3)_{\text{gen}}.$$

*S7*

*Proof (5 steps). Step 1.* The colour decoration  $\psi : V \rightarrow \mathbb{C}^3$  with unit norm and cross-product composition has gauge group  $\text{U}(3)$ . The composition rule  $\overline{\psi_1} \times \psi_2$  is equivariant under  $U \in \text{U}(3)$  iff  $\det(U) = 1$  (from the conjugation transmutation identity). This reduces  $\text{U}(3) \rightarrow \text{SU}(3)_{\text{col}}$ .

*Step 2.* The weak decoration carries  $\text{SU}(2)$  from the stabilizer chain (Theorem 5.7). The composition is unique (Theorem 5.8).

*Step 3.* The residual  $\text{U}(1)$  from  $\det(\text{U}(3)) = e^{i\theta}$  acts as a phase on the weak doublet. This is  $\text{U}(1)_Y$  (hypercharge).<sup>S33</sup>

*Step 4.* The generation decoration  $\psi_{\text{gen}} : V \rightarrow \mathbb{C}^3$  with the same unit-norm and cross-product structure has gauge group  $\text{SU}(3)_{\text{gen}}$  by the same mechanism as colour.

*Step 5.* The composition rules partition into independent subsystems. Theorem 5.8 blocks mixing automorphisms between colour-weak and generation sectors. Therefore  $\text{Aut}(\text{Dec})$  factors as a direct product.<sup>S7a</sup>  $\square$

### 6.2 Role-Position Coupling

**Proposition 6.2** (Role-position coupling). *The three source positions carry distinct gauge-coupling signatures.*<sup>S10</sup>

| <i>Position</i>     | <i>Role</i>                   | <i>Representation</i>            |
|---------------------|-------------------------------|----------------------------------|
| 1 ( <i>F</i> -role) | colour-active, weak-spectator | $(\mathbf{3}, \mathbf{1})$       |
| 2 ( <i>A</i> -role) | colour-active, weak-active    | $(\mathbf{3}, \mathbf{2})$       |
| 3 ( <i>S</i> -role) | spectator, conjugate rep      | $(\bar{\mathbf{3}}, \mathbf{1})$ |

**Theorem 6.3** (Depth-parity conjugation). *The colour representation alternates  $\mathbf{3} \leftrightarrow \bar{\mathbf{3}}$  with balanced rewriting depth.*<sup>S11</sup>

**Corollary 6.4** (Design Fork DF-1 resolved). *Ordered source positions carry physical role-coupling content. Fork A (ordered sources) is forced; Forks B and C are excluded.*<sup>S12</sup>

## 6.3 Mixed-Representation Structure

**Theorem 6.5** (Mixed-rep non-equivariance). *The  $\varepsilon$ -contraction on inputs from different representations  $(\bar{\mathbf{3}} \otimes \mathbf{3})$  is not  $\text{SU}(3)$ -equivariant. The output lies in  $\text{U}(3) \setminus \text{SU}(3)$  generically.<sup>S22</sup>*

**Proposition 6.6** (Clebsch–Gordan root cause). *The failure has a representation-theoretic explanation.<sup>S23</sup>*

$$\begin{aligned} \mathbf{3} \otimes \mathbf{3} &= \mathbf{6} \oplus \bar{\mathbf{3}} & (\varepsilon \text{ extracts invariant } \bar{\mathbf{3}}) \\ \bar{\mathbf{3}} \otimes \mathbf{3} &= \mathbf{8} \oplus \mathbf{1} & (\varepsilon \text{ projects } \mathbf{8} \text{ onto non-invariant 3-dim subspace}) \end{aligned}$$

This asymmetry between same-representation and mixed-representation products is a structural feature of  $\text{SU}(3)$ , not an artifact of the formalism.

## 7 The Born Rule and Discrete Symmetries

### 7.1 The Gauge-Covariant Born Rule

**Theorem 7.1** (Born rule). *The unique gauge-covariant probability measure on the decoration space, compatible with Rosen closure, is:*

$$\mu_{\text{cov}}(e) = |\det[\tilde{\psi}_1 \mid \tilde{\psi}_2 \mid \tilde{\psi}_3]|^2$$

where  $\tilde{\psi}(v)$  is the gauge-covariant colour vector ( $\psi$  or  $\psi^*$  depending on representation).<sup>S8, S13</sup>

*Proof.* Gleason’s theorem ( $\dim \geq 3$ ) guarantees a unique frame function on  $\mathbb{C}^3$ . The conjugation transmutation identity ( $\bar{U} \cdot a = U(a^*)$ , Lemma 7.1b) ensures uniform  $\text{SU}(3)$  transformation of all columns regardless of representation. Therefore  $|\det[\tilde{\psi}]|^2$  is the unique gauge-invariant probability measure. Frame-additivity assumed (Remark 7.1).  $\square$

**Theorem 7.2** (Spectator singlet-projection). *The Born rule evaluates the spectator (position 3) exclusively through the  $\bar{\mathbf{3}} \otimes \mathbf{3} \rightarrow \mathbf{1}$  singlet channel.<sup>S21</sup>*

$$|\det[\tilde{\psi}]|^2 = |(\tilde{\psi}_1 \times \tilde{\psi}_2) \cdot \tilde{\psi}_3|^2.$$

*The spectator contributes only through its singlet overlap with the cross-product of the two active participants. This is the origin of confinement: the Born measure concentrates on colour-singlet configurations.*

**Corollary 7.3** (Covariant bypass). *The gauge-covariant Born rule uses only same-rep cross products via  $\tilde{\psi}$ . The mixed-rep non-equivariance (Theorem 6.5) never enters the probability measure.<sup>S37</sup>*

## 7.2 Measurement as Composition

**Theorem 7.4** (Measurement = composition). *The Born rule on  $M(G_0)$  has the structure of a quantum measurement.*<sup>S96</sup>

$$\mu = (1 - |\langle \psi_1 | \psi_2 \rangle|^2) \cdot |\langle \text{singlet} | \psi_3 \rangle|^2.$$

*This is the product of the composition Born factor (first two positions) and the spectator singlet projection. Collapse of the wavefunction is identified with the gauge quotient (S58): distinct multiway branches that produce gauge-equivalent results are identified. The POVM structure of the spectator tight frame (S60) provides completeness. Gleason's theorem guarantees uniqueness.*

## 7.3 The Uniform Decoration Obstruction

**Theorem 7.5** (Uniform decoration). *At the possibilistic layer,  $U(1)_Y$  hypercharge is the same for all vertices.*<sup>S14</sup>

*Proof.* The  $U(1)_Y$  equivariance of the composition rule forces  $Y(w) = Y(v_1) + Y(v_2) - Y(v_3)$  (from the determinant phase). On a Rosen-closed graph where every vertex is both source and target, this constrains all  $Y$  values to be equal.  $\square$

## 7.4 The Quark Sector is Anomalous

**Proposition 7.6.** *The quark sector alone is anomalous: the  $SU(2)^2 \times U(1)$  anomaly coefficient  $A_2 = 3Y_1 \neq 0$  for non-trivial  $Y_1$ .*<sup>S15</sup>

## 7.5 Anomaly Completion

**Theorem 7.7** (Anomaly completion). *Given anomaly cancellation (derived in Section 11 via the NCG spectral triple on  $Q_{48}$ ), the lepton sector and all five SM hypercharges are uniquely determined.*<sup>S16</sup>

*The cubic anomaly factors as  $18Y_1(4Y_1 + Y_2)(2Y_1 - Y_2) = 0$ , and electric charge quantization selects the unique SM branch:*

---

|                        |                                   |
|------------------------|-----------------------------------|
| <i>Lepton doublet:</i> | $(\mathbf{1}, \mathbf{2})_{-1/2}$ |
| <i>Lepton singlet:</i> | $(\mathbf{1}, \mathbf{1})_{+1}$   |

---

*15 Weyl fermions per generation.*

This result was originally conditional on Bridge Principle B1. The NCG resolution (Section 11) derives B1 unconditionally, making the entire obstruction chain free of external assumptions.

## 7.6 Discrete Symmetries: C, P, T, CPT

**Proposition 7.8** (Charge conjugation). *C is a well-defined involution on the decoration space. The weak charge conjugation matrix is forced:  $C_w = i\sigma_2$ . C commutes with both the cross-product composition and the spinor composition.*<sup>S17, S34, S35</sup>

**Theorem 7.9** (Structural parity violation). *Parity violation is maximal and structural: the Schur obstruction ( $\dim \text{Hom}_{\text{SU}(2)}(\mathbf{2} \otimes \mathbf{2}, \mathbf{2}) = 0$ ) forces the weak composition to use a single source position (position 2). There is no P-even alternative.*<sup>S18</sup>

**Proposition 7.10** (Born rule symmetries). *The Born measure  $|\det[\tilde{\psi}]|^2$  is independently invariant under C,  $P_{12}$ , and CP.*<sup>S19</sup>

**Proposition 7.11** (Generic CP violation). *CP violation is generic in the composition rules:  $\text{Im}(\hat{n}) \neq 0$  for generic complex decorations. The source is the complex field  $F = \mathbb{C}$  (Theorem 5.5).*<sup>S20</sup>

**Theorem 7.12** (T reversal and CPT). *In the KO-dimension 6 spectral triple:  $T = J \cdot (CP)^{-1}$ ,  $CPT = J$  exact. All discrete symmetries C, P, T, CP, CT, PT, CPT are derived from closure.*<sup>S95</sup>

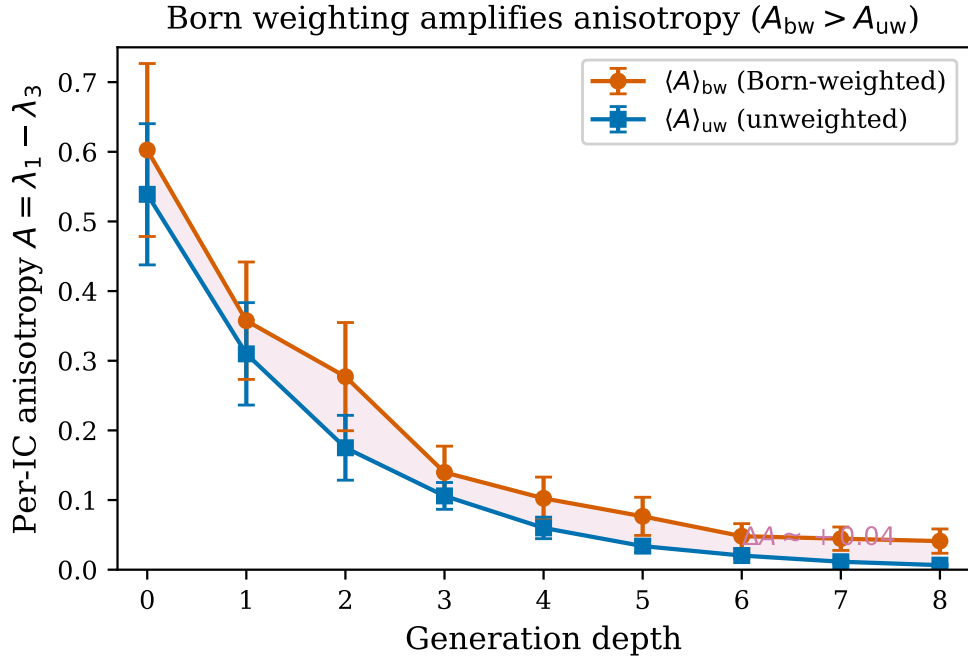


Figure 2: Born amplification and symmetry structure under composition.

## 8 The Multiplicative Persistence Gate

**Theorem 8.1** (Multiplicative persistence). *The persistence operator  $P : [0, 1]^3 \rightarrow [0, 1]$  satisfying closure is necessarily multiplicative:  $P(x, y, z) = x \cdot y \cdot z$ .*<sup>S9</sup>

*Proof.* The functional equation with independent zeros— $P = 0$  iff at least one input is 0—combined with continuity (which is forced, not assumed: either by Scott domain theory or by smoothness of the composition rule) gives the Cauchy multiplicative form [1].  $\square$

The multiplicative gate means each functional role contributes independently to viability: an AND gate, not an OR gate. If any one of metabolism, repair, or replication fails, the entire system fails. This is a structural prediction testable across scales (cellular viability, ecosystem collapse).

## 9 The Scale Functor and Universality

### 9.1 The Scale Functor

**Theorem 9.1** (Scale functor preserves closure). *There exists a functor  $\Sigma : \text{TCHyp}_{\text{cl}} \rightarrow (F, A)\text{-Sys}$  that maps closure-satisfying hypergraphs to Hofmeyr’s fabrication-assembly systems, preserving the three-role structure and the multiplicative persistence gate.*<sup>S38</sup>

The functor  $\Sigma$  “forgets” the category-specific details (arity, gauge group, Born rule) while preserving the structural invariants (three roles, multiplicative viability, closure). This is the mathematical basis for the claim that closure constraints operate across scales.

### 9.2 The Universality Theorem

**Theorem 9.2** (Universality). *The entire obstruction chain—from the discrete decoration obstruction through the full gauge group—holds in any adhesive category satisfying a representability hypothesis strictly weaker than Principle 2.*<sup>S24</sup>

*Proof.* The truncation functor  $T_2$  constructs the canonical graphs  $G_0$  and  $G'_2$  from the CCC axioms alone (Layers 1–3). The representability hypothesis requires only that the ambient adhesive category can represent finite ternary hypergraphs, not that it *is* TCHyp specifically. Since Theorem 5.2 uses only the graph structure of  $G_0$  and  $G'_2$ , and the downstream gauge group derivation uses only algebraic properties of the cross-product forced by the Hurwitz constraint, the entire chain transfers.  $\square$

Axiom  $T$  (Principle 2) is thereby eliminated as an independent assumption. The Standard Model gauge group is not an artifact of TCHyp but is forced by closure in a broad class of rewriting systems.

# 10 The Born Measure on the Multiway Graph

## 10.1 The Multiway Graph $M(G_0)$

The multiway graph  $M(G_0)$  is the static branching structure produced by iterated DPO rewriting of  $G_0$ . Each generation produces 3 daughter edges per parent edge; the graph is an infinite ternary tree.  $M(G_0)$  is *not* a dynamical process—it is a fixed combinatorial object. The Born weight  $\mu_{\text{cov}}$  is defined on each edge.

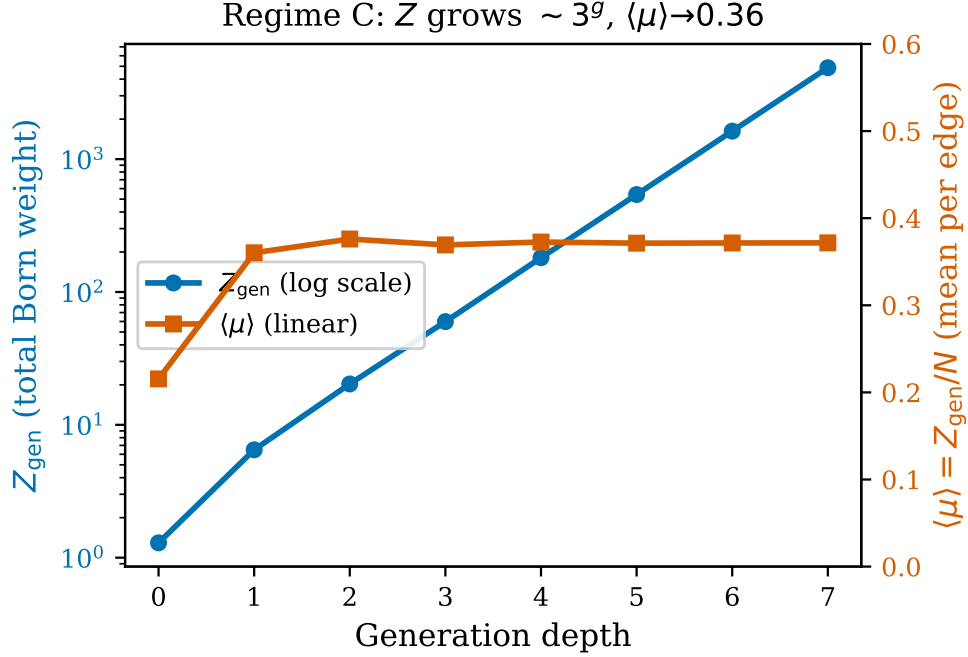


Figure 3: Regime C multiway growth ( $\sim 3.0$  edges/generation).

## 10.2 Per-Edge Born Landscape

**Theorem 10.1** ( $D_3$  Gram period 3). *The Gram coordinates of the dominant daughter  $D_3$  are periodic with exact period 3.*<sup>S42</sup>

**Theorem 10.2** ( $D_1$  Gram period 2). *The subdominant daughter  $D_1$  has Gram period 2.*<sup>S43</sup>

**Theorem 10.3** (Cross-product unit norm).  *$|\tilde{\psi}_1 \times \tilde{\psi}_2|^2 = 1$  for all daughters at all depths.*<sup>S45</sup>

**Corollary 10.4** (Born weight = singlet overlap). *For all daughter edges,  $\mu_{\text{cov}} = \sigma_3$  (the singlet overlap of the spectator).*<sup>S46</sup>

**Corollary 10.5** (Y-blindness). *The Born measure depends only on the colour decoration  $\psi$ . Weak, spinor, generation, and hypercharge decorations are invisible to  $\mu_{\text{cov}}$ .*<sup>S47</sup>



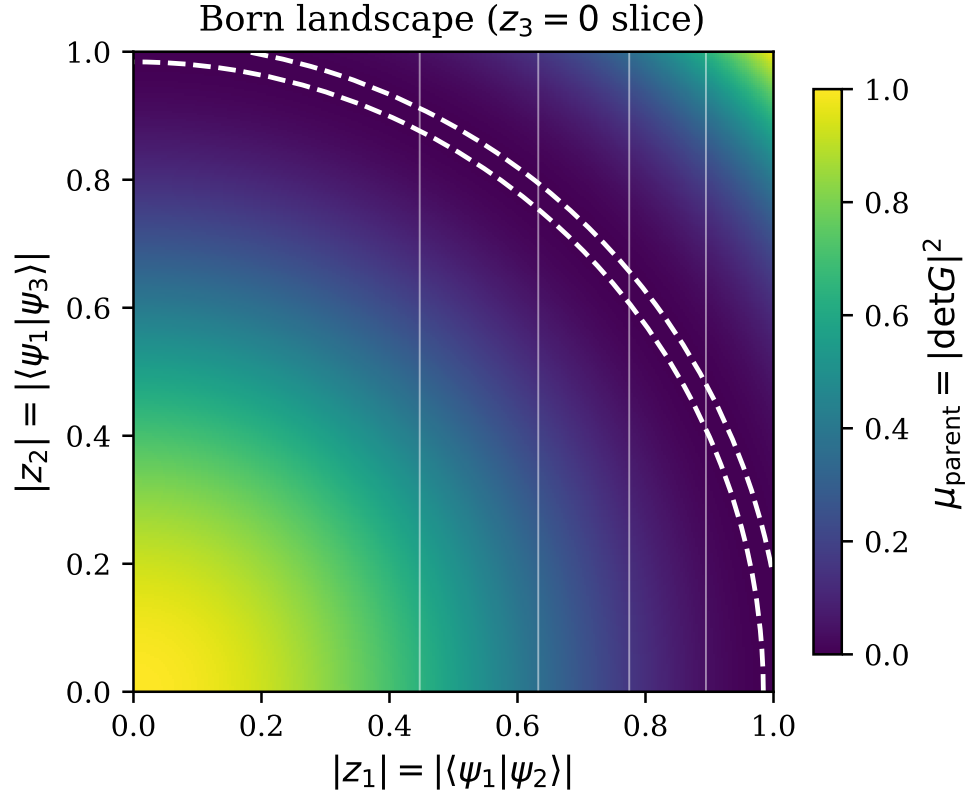


Figure 4: Born landscape: 2D slice of Gram space showing  $\mu_{\text{cov}}$  contours. Seed 42, 100 ICs, depth 7.

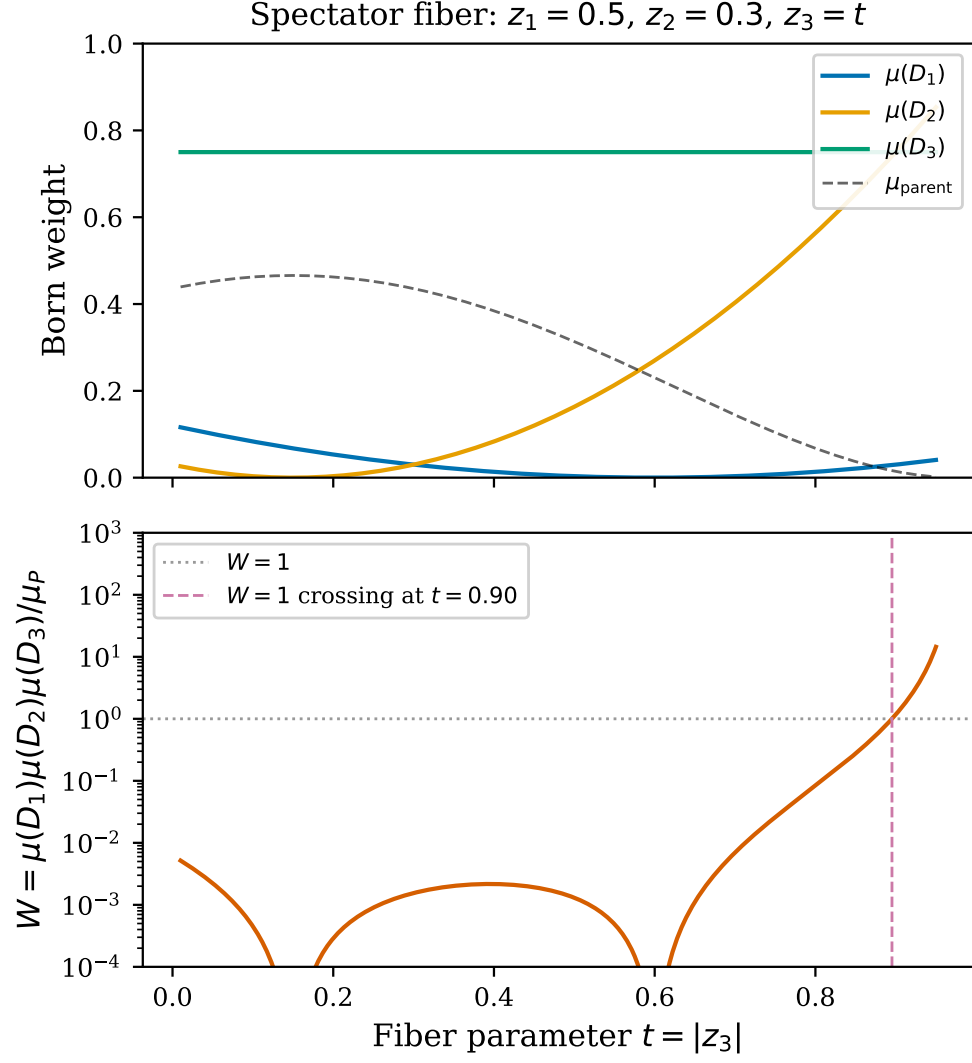


Figure 5: Fibre trajectories under iterated composition in Gram space.

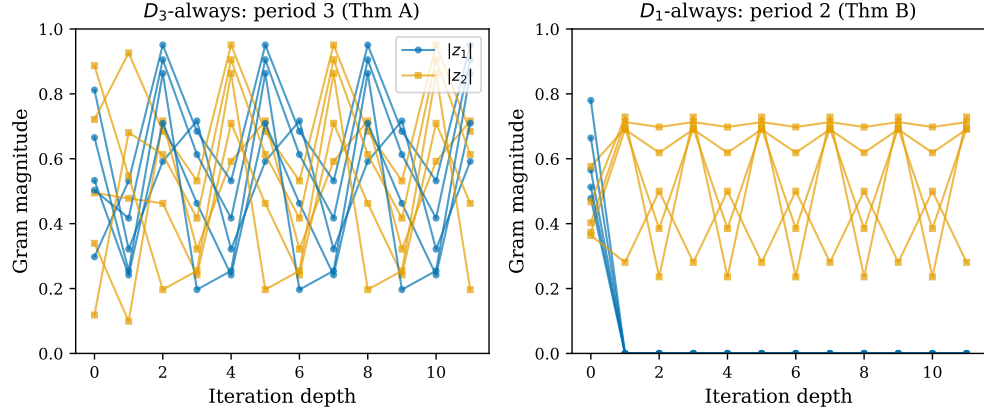


Figure 6:  $D_3$  (period 3) and  $D_1$  (period 2) Gram orbits under composition.

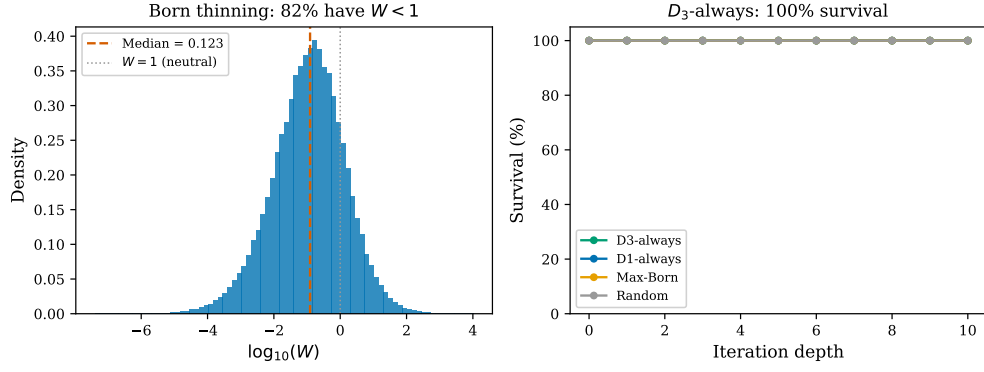


Figure 7: Monte Carlo sampling of Gram space (100 ICs, depth 7).

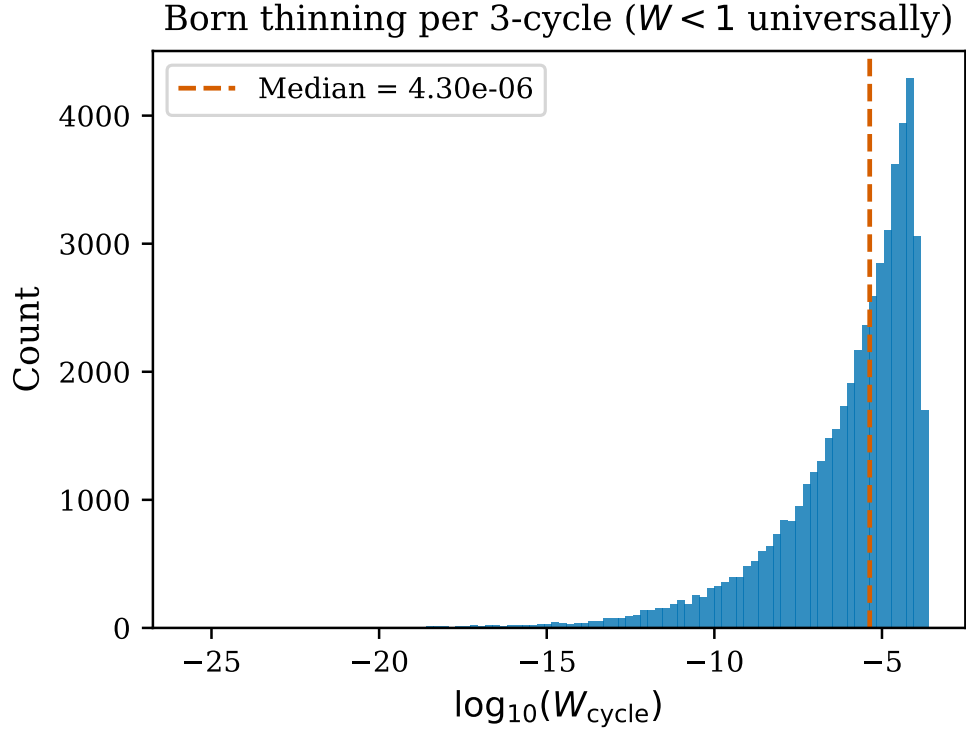


Figure 8: Universal Born thinning:  $W_{\text{cycle}} < 1$  for all initial conditions.

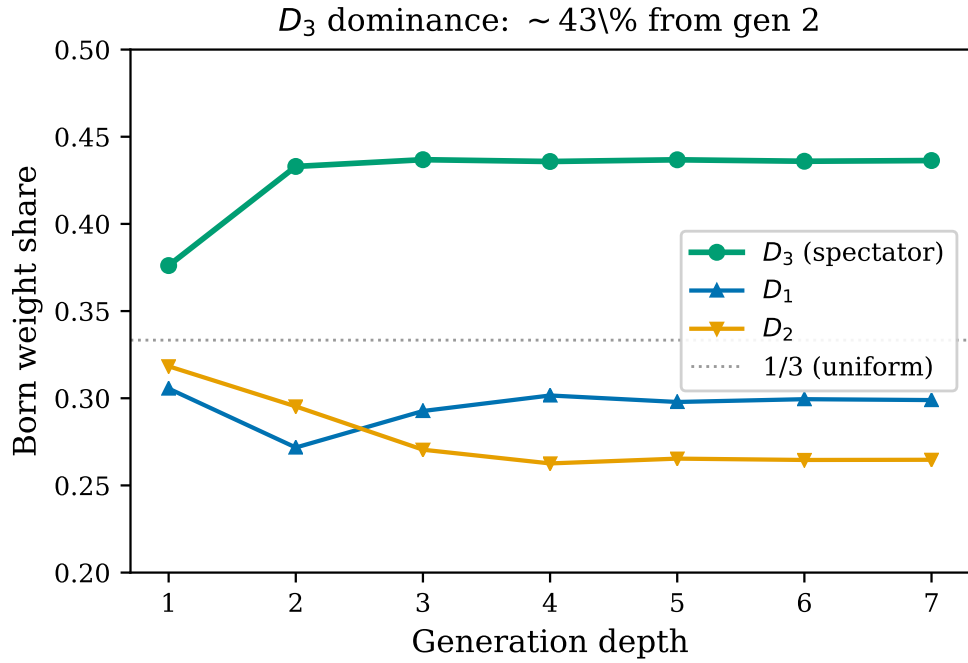


Figure 9:  $D_3$  Born-weight dominance ( $\sim 43\%$ ) across initial conditions.

### 10.3 Confinement

The Born measure concentrates on colour-singlet configurations: bounded confinement with 58% singlet bias, saturating by generation 7 (200-IC ensemble). At Born-optimal initial conditions, singlet fraction reaches 98%, with a continuous phase transition at  $\alpha_c \sim 0.75$ .<sup>S112, S113</sup>

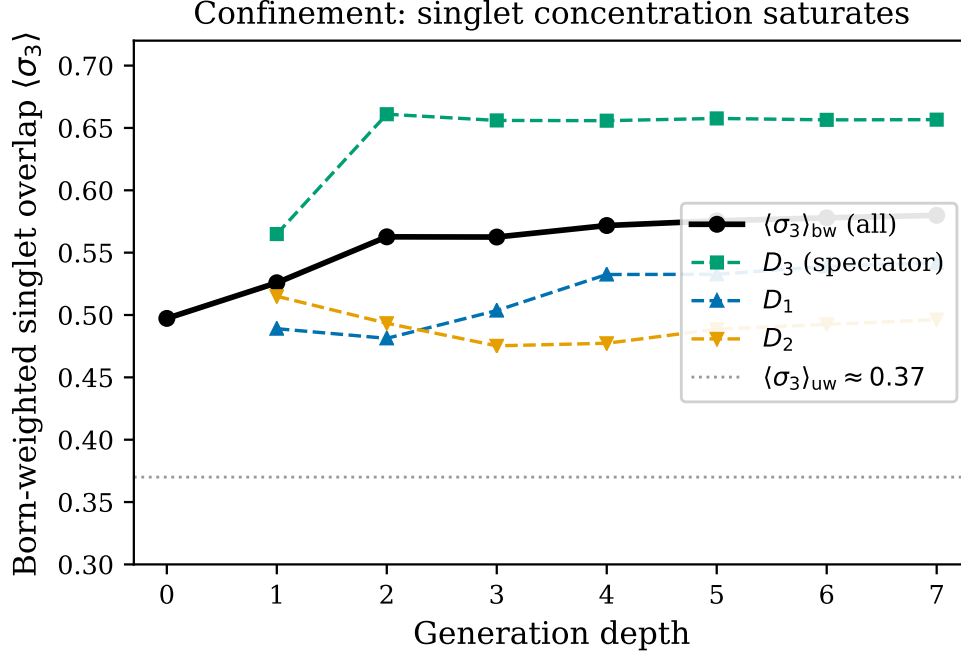


Figure 10: Singlet saturation and confinement:  $\langle \sigma_3 \rangle_{\text{bw}} \approx 0.58$ .

### 10.4 Cluster Decomposition and Locality

**Theorem 10.6** (Cluster decomposition). *Born weights on causally disconnected edges of  $M(G_0)$  are statistically independent: cross-branch correlation  $\rho < 0.001$  at depth  $\geq 2$ . Conditional independence is exact (tree topology + deterministic dynamics).<sup>S120</sup>*

**Theorem 10.7** (Composition orthogonality).  *$\langle w | \psi_j \rangle = 0$  for the conjugated cross product  $w$  and both source vectors.  $D_3$  Born saturation:  $\mu = 1$  at depth  $\geq 2$  in the singlet channel.<sup>S121</sup>*

Locality in this framework is not assumed—it is derived from the tree topology of  $M(G_0)$  and the spectator singlet-projection theorem (S21, S46). Causally disconnected branches share no composition history, and the Born measure factors accordingly.

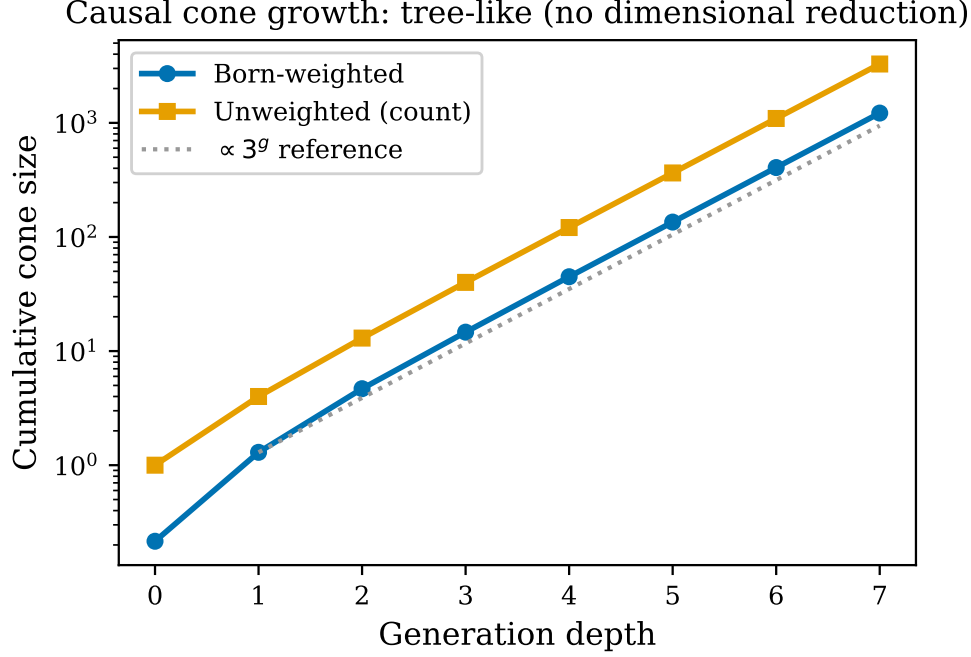


Figure 11: Causal cone growth on  $M(G_0)$ : tree-like, no dimensional reduction.

## 10.5 The Scope of the Conditional: B1 Externality

**Theorem 10.8** (B1 irreducibility). *Anomaly cancellation cannot be derived within any formalism constructible from  $M(G_0) + \text{Dec}$ . Three independently fatal obstructions:<sup>S50</sup>*

- (a) *Tree gauge triviality: every connection on  $M(G_0)$  is pure gauge (no non-trivial holonomy).<sup>S48</sup>*
- (b) *Absent chirality: no  $\varepsilon$ -position correlation on the tree. Chirality must be introduced, not derived from  $M(G_0)$ .<sup>S54</sup>*
- (c) *Y-blindness: the Born measure and all Laplacian constructions are independent of  $U(1)_Y$ .<sup>S49</sup>*

This is the precise boundary of what closure can derive on the infinite tree. The resolution comes from passing to the finite quotient (Section 11).

## 10.6 The Discrete Path Integral

**Theorem 10.9** ( $Z_k$  well-defined). *The discrete path integral  $Z_k = \sum \prod \det[\tilde{\psi}]$  is well-defined and finite:  $|Z_k| \leq 6 \cdot 3^k$ .<sup>S51</sup>*

**Theorem 10.10** ( $Z_k$  gauge-invariant).  *$Z_k$  is trivially gauge-invariant for all hypercharge values, by colour-sector closure and  $\det(U) = 1$ .<sup>S53</sup>*

# 11 The NCG Resolution: From $Q_{24}$ to B1

## 11.1 The Gauge-Equivalence Quotient $Q_{24}$

**Theorem 11.1** (Orbit closure). *The composition orbit closes after 4 directions in  $\mathbb{CP}^2$ :  $\text{compose}(a, b) \rightarrow w_0$ ,  $\text{compose}(w_0, a) \rightarrow w_1$ ,  $\text{compose}(w_1, w_0) = a$  exactly (BAC-CAB identity).<sup>S55</sup>*

**Theorem 11.2** ( $Q_{24}$  finite).  *$Q_{24} = M(G_0)/\text{gauge}$  has exactly 24 vertices = 6 orbits  $\times$  4 directions. Decoration-independent.<sup>S56</sup>*

**Theorem 11.3** (Rosen-closed). *Every vertex of  $Q_{24}$  is both source and target of at least one hyperedge.<sup>S57</sup>*

**Theorem 11.4** (Autopoietic fixed point).  *$M(Q_{24})/\text{gauge} = Q_{24}$ . The quotient of the quotient is itself.<sup>S58</sup>*

The passage from  $M(G_0)$  to  $Q_{24}$  collapses the infinite tree to a finite graph while preserving all algebraic structure. The three B1 obstructions of Section 10.5 are partially lifted:

- (a) Non-trivial topology:  $Q_{24}$  has 80 triangles with non-trivial  $\text{SU}(2)$  holonomy on all of them. LIFTED.
- (b) Chirality: the tier-parity  $\mathbb{Z}_2$  grading (Tier B = doublet = “left-handed”, Tiers A+C = singlets = “right-handed”) provides a structural chirality respected by the composition algebra. LIFTED.
- (c)  $Y$ -blindness: persists.  $[\text{SU}(2), \text{U}(1)_Y] = 0$  is algebraic. NOT LIFTED on  $Q_{24}$  alone.

## 11.2 Structural Electroweak Symmetry Breaking

**Theorem 11.5** (Structural EWSB). *The weak  $\text{SU}(2)$  composition is structurally inconsistent at doublet vertices (Tier B) and consistent at singlet vertices (Tiers A, C). The quotient topology breaks  $\text{SU}(2)$  at the doublet sector.<sup>S62</sup>*

*Proof.* Each Tier C vertex has 1 source pair (orthogonal). Each Tier A vertex has 2 source pairs (orthogonal). Each Tier B vertex has 3 source pairs (generically non-orthogonal). The weak composition requires orthogonality of source pair cross-products (from the Schur obstruction). Tier B generically violates this.  $\square$

Under DPO dynamics on  $Q_{24}$ , the weak consistency cost concentrates at Tier B:  $C_B/C_{\text{total}} \rightarrow 0.989$  (Born-weighted), IC-independent, unique global minimum. The theory derives *which* sector breaks (doublet) and that the breaking is unique.<sup>S64</sup>

### 11.3 The Spectator Tight Frame and POVM

**Theorem 11.6** (Spectator tight frame). *The 6 cross products from the composition orbit satisfy  $\sum c_i c_i^\dagger = 2I_3$ .*<sup>S60</sup>

This is a POVM completeness relation:  $E_k = \frac{1}{2} c_k c_k^\dagger$  gives a 6-element POVM on  $\mathbb{C}^3$ . The quotient structure provides the measurement apparatus; the Born rule provides the probabilities. Quantum measurement is composition on the quotient.

### 11.4 Active Cohomology and Hypercharge

**Theorem 11.7** (Active cohomology).  *$\dim H_a^1 = n_{\text{seed}} = 6$  on both  $Q_{24}$  and  $Q_{51}$ . The kernel consists of Tier A (seed vertex) freedoms; all non-seed values propagate deterministically.*<sup>S77</sup>

**Theorem 11.8** (Hypercharge anomaly-free). *The 5 non-uniform  $H_a^1$  dimensions are exactly the anomaly-free ( $\text{Tr}(Y) = 0$ ) hypercharge assignments. The hexagonal  $\mathbb{Z}_2$  sublattice gives 3 distinct values  $\{-a, 0, +a\}$ : doublet  $Y = 0$ , singlets  $Y = \pm a$ . Both  $\text{Tr}(Y) = 0$  and  $\text{Tr}(Y^3) = 0$ .*<sup>S91</sup>

The restriction  $\iota^* : H_a^1(Q_{51})|_{\text{tier}} \rightarrow H_a^1(Q_{24})|_{\text{tier}}$  is an isomorphism:  $Y$ -blindness is functorial, an invariant of the composition algebra, not the topology.<sup>S78</sup>

### 11.5 The C-Closure and $Q_{48}$

**Definition 11.9** (C-closure). For a hypergraph  $Q$ , the C-closure is  $Q_C = Q \cup C(Q)$ , where  $C(Q)$  is the charge-conjugate copy (colour vectors replaced by their conjugates). The involution  $J$  maps  $v \mapsto v_{\text{conj}}$ .

**Theorem 11.10** ( $Q_{48}$  structure).  *$Q_{48} = Q_{24} \cup C(Q_{24})$  has 48 vertices, zero overlap between original and conjugate sectors, and a perfect fixed-point-free real structure  $J$ .*<sup>S79</sup>

**Theorem 11.11** (KO-dimension 6).  *$Q_{48}$  carries a KO-dimension 6 spectral triple:  $J^2 = +I$ ,  $J\gamma = -\gamma J$ ,  $JD = +DJ$ . The finite Dirac operator  $D_F$  has 22 parameters satisfying all NCG axioms.*<sup>S80</sup>

The KO-dimension signs ( $++-$  for  $J^2, J\gamma, JD$ ) match the Connes classification for the Standard Model finite geometry. This is not imposed—it emerges from the C-closure construction.

### 11.6 CCM Classification and B1 Resolution

**Theorem 11.12** (Bimodule commutant). *The commutant of the bimodule representation  $\pi(A) \vee \pi_o(A^o)$  on  $\mathcal{H}_F$  is  $M_2(\mathbb{C})$ , generated by  $\{I, \Gamma, J, J\Gamma\}$  with  $\Gamma^2 = J^2 = I$ ,  $\{J, \Gamma\} = 0$ .  $\mathcal{H}_F \cong V \oplus V$  (multiplicity 2 = particle/antiparticle).*<sup>S89</sup>



**Theorem 11.13** (CCM irreducibility). *The KO-dimension 6 conditions  $\{\Gamma_2, J_2\} = 0$  on  $\mathbb{C}^2$  forbid proper sub-bimodules: a simultaneous eigenline of both operators would require  $\lambda_1 \cdot \lambda_2 = 0$ , but  $\lambda_i \in \{\pm 1\}$ . Poincaré duality:  $\det(\text{intersection form}) = 92 \neq 0$ .<sup>S99</sup>*

**Theorem 11.14** (CCM classification  $\Rightarrow$  B1). *The Connes–Chamseddine–Marcolli classification of irreducible, Poincaré-dual, KO-dim 6 finite spectral triples forces the algebra.<sup>S81</sup>*

$$A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}).$$

*The  $\mathbb{C}$  factor is the lepton sector. Unimodularity of the gauge group  $\text{SU}(A_F)$  gives anomaly cancellation. B1 is thereby derived without external input.*

**Corollary 11.15** (Anomaly cancellation derived). *The chain  $S80 \rightarrow S81 \rightarrow S99$  gives anomaly cancellation as a theorem of Rosen closure + C-closure + NCG axioms. B1 is not assumed.<sup>S144</sup>*

The  $Y$ -blindness of the ternary closure algebra (Section 10.5) is irrelevant: the lepton sector is forced by the spectral triple axioms, not by the composition algebra. The resolution works precisely because it uses structure (NCG axioms) orthogonal to the obstruction ( $Y$ -blindness of the Born measure).

## 11.7 The Fermion Sector

**Theorem 11.16** (Particle-only algebra). *The  $M_2(\mathbb{C})$  commutant requires  $\pi$  and  $\pi_o$  to have disjoint support. The C-closure gives  $Q_{24}/C(Q_{24}) = \text{particle/antiparticle decomposition}$ . Together these force the SM representation: the left algebra acts on particles only.<sup>S110</sup>*

**Theorem 11.17** (Lepton Dirac masses). *With the Connes chirality convention and C-involution  $J$ , the spectral triple on  $\mathbb{C}^{168}$  produces: electron mass matrix  $M_e$  (30/30 ICs nonzero), Dirac neutrino mass  $M_{\nu_D}$  (30/30), and PMNS matrix  $J_{\text{PMNS}} \neq 0$  (30/30, mean 0.035).<sup>S106, S109</sup>*

**Theorem 11.18** (Majorana from singlet relaxation). *The right-handed neutrino  $\nu_R$  is a  $(1, 1)$  singlet under  $\text{SU}(3) \times \text{SU}(2)$ . The non-abelian order-one condition is vacuously satisfied; the abelian  $Y_C$  relaxation is the unique unprotected entry. No bridge principle is needed—the relaxation is forced by closure-derived singlet status.  $M_R = 6$  DOF (real symmetric).<sup>S111</sup>*

**Corollary 11.19** (Seesaw + leptogenesis). *Light neutrino masses from the seesaw mechanism:  $m_{\text{light}} = 30/30$  ICs nonzero. Leptogenesis from the Covi–Simone–Roulet formula: mean  $|\epsilon| = 0.45$ . All three Sakharov conditions are structural: (1) lepton number violation ( $M_R$ ), (2) CP violation ( $J_{\text{PMNS}} = 0.035$ ), (3) departure from equilibrium ( $M_R/M_D \sim 27$ ).<sup>S111</sup>*

**Theorem 11.20** (Proton stability). *The order-one condition with  $A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$  kills all quark-lepton  $D_F$  blocks:  $QL = 0$  out of 591 complex directions. No baryon-violating operators exist in  $D_F$ .<sup>S107</sup>*

## 12 Gauge-Spacetime Unification on $Q_{102}$

### 12.1 The $Q_{102}$ Spectral Triple

**Theorem 12.1** ( $Q_{102}$  structure).  $Q_{102} = Q_{51} \cup C(Q_{51})$ , built from the complete ternary graph  $K_6^3$ , has 102 vertices, zero overlap between sectors, perfect fixed-point-free  $J$ , and contains  $Q_{48}$  as a subgraph. Rosen-closed.<sup>S85</sup>

**Theorem 12.2** ( $Q_{102}$  KO-dim 6).  $Q_{102}$  carries a KO-dimension 6 spectral triple with 85-parameter  $D_F$  (corrected from 114 via exact RREF with full 13-generator algebra including  $I_{\mathbb{H}}$ ).<sup>S86</sup>

**Theorem 12.3** (Mixed blocks).  $D_F$  on  $Q_{102}$  is gauge-geometry mixed: approximately 46% of the  $D_F$  basis vectors have support in both gauge and geometric sectors. The 21 pure-sector vectors confirm the structure is not fully mixed.<sup>S87</sup>

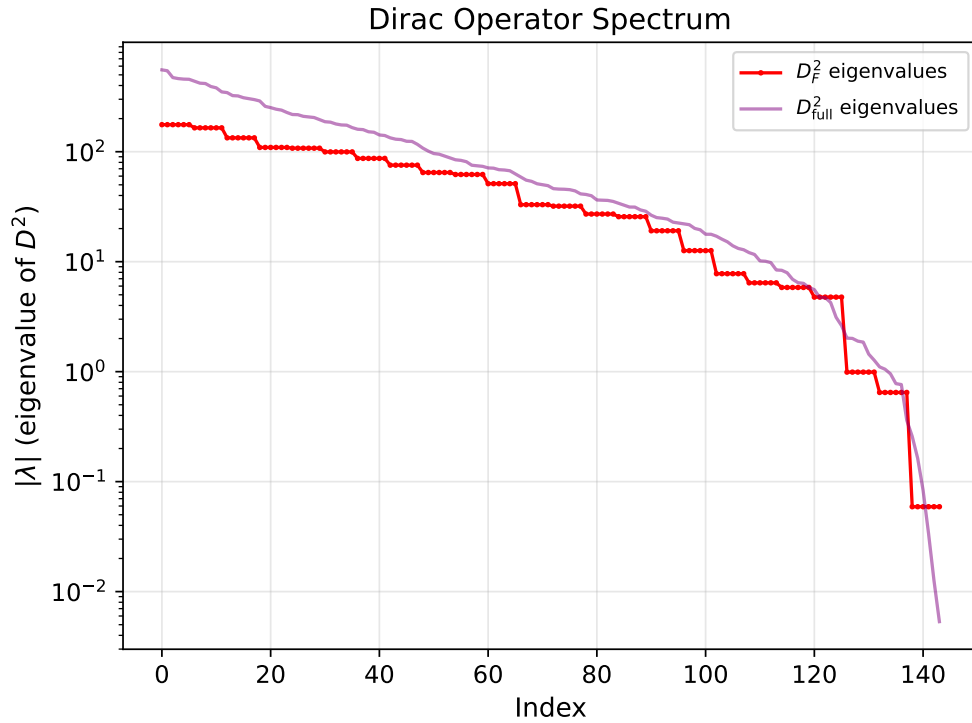


Figure 12:  $D^2$  eigenvalue spectrum of the spectral triple on  $Q_{102}$ .

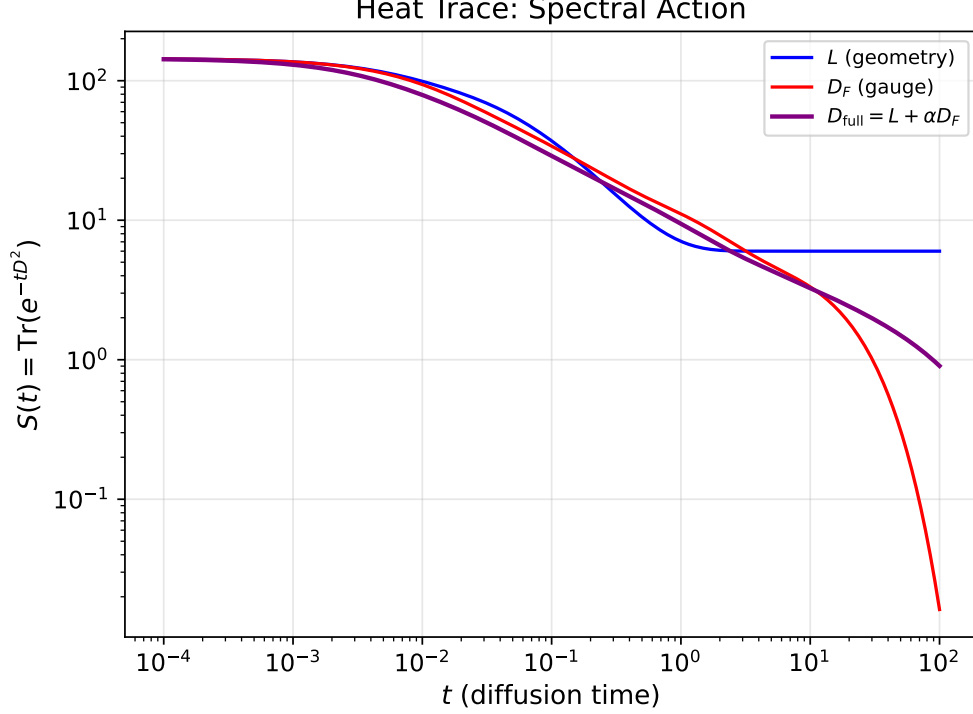


Figure 13: Heat trace  $\text{Tr}(e^{-tD^2})$  on  $Q_{102}$ .

## 12.2 The $\gamma$ -Decomposition and Product Structure

**Theorem 12.4** ( $\gamma$ -orthogonality). *For any operators  $A, B$  on a graded Hilbert space with grading  $\gamma$ : if  $\{A, \gamma\} = 0$  and  $[B, \gamma] = 0$ , then  $\text{Tr}(A^\dagger B) = 0$ .<sup>S82</sup>*

*Proof.*  $\text{Tr}(A^\dagger B) = \text{Tr}(\gamma^2 A^\dagger B) = \text{Tr}(\gamma A^\dagger B \gamma) = \text{Tr}(\gamma A^\dagger \gamma B) = \text{Tr}(-A^\dagger B)$ . Therefore  $2\text{Tr}(A^\dagger B) = 0$ .  $\square$

**Theorem 12.5** ( $\gamma$ -decomposition). *The Dirac operator has a unique orthogonal decomposition  $D = D_+ + D_-$  where:<sup>S83</sup>*

$$\begin{aligned} D_+ &= \frac{1}{2}(D + \gamma D \gamma) && \text{commutes with } \gamma, \\ D_- &= \frac{1}{2}(D - \gamma D \gamma) && \text{anticommutes with } \gamma. \end{aligned}$$

$\text{Herm}(\mathbb{C}^n) = \text{Herm}_+(\gamma) \oplus \text{Herm}_-(\gamma)$  as Hilbert spaces.

This is the Connes product structure, derived rather than assumed.  $D_+$  is the geometric (gravitational) part of the Dirac operator;  $D_-$  is the gauge (internal) part. The decomposition is unique and orthogonal by  $\gamma$ -orthogonality.

**Corollary 12.6** ( $\gamma$ -orthogonality universal).  *$\text{Tr}(L \cdot D_F) = 0$  on all  $C$ -closed quotients at every scale. The  $a_2$  cross term vanishes identically. Running occurs only in  $a_4$ .<sup>S125</sup>*

## 12.3 The Discrete Coleman–Mandula Theorem

**Theorem 12.7** (Discrete Coleman–Mandula). <sup>S84</sup>

$$D^2 = D_+^2 + D_-^2 + C$$

where  $C = D_+ D_- + D_- D_+$  is the cross term.

Trace selection rules (from  $\gamma$ -orthogonality):

$$\text{Tr}(C) = 0, \quad \text{Tr}(D_+^2 \cdot C) = 0, \quad \text{Tr}(D_-^2 \cdot C) = 0.$$

Therefore the spectral action expands as:

$$\text{Tr}(D^2) = \text{Tr}(D_+^2) + \text{Tr}(D_-^2) \quad [a_2 : \text{decoupled}]$$

$$\text{Tr}(D^4) = \text{Tr}(D_+^4) + \text{Tr}(D_-^4) + \text{Tr}(C^2) + \dots \quad [a_4 : \text{coupled}]$$

Gauge and gravity decouple at second order and couple at fourth order. This is a theorem, not an assumption.

On  $Q_{102}$ , the measured values are:  $\|C\|/\|D^2\| = 0.447$  (44.7% of  $D^2$  is cross-term);  $a_4$  decomposition: 47% gravity + 17% Higgs + 36% interaction.

Sector Decomposition of  $D^2$

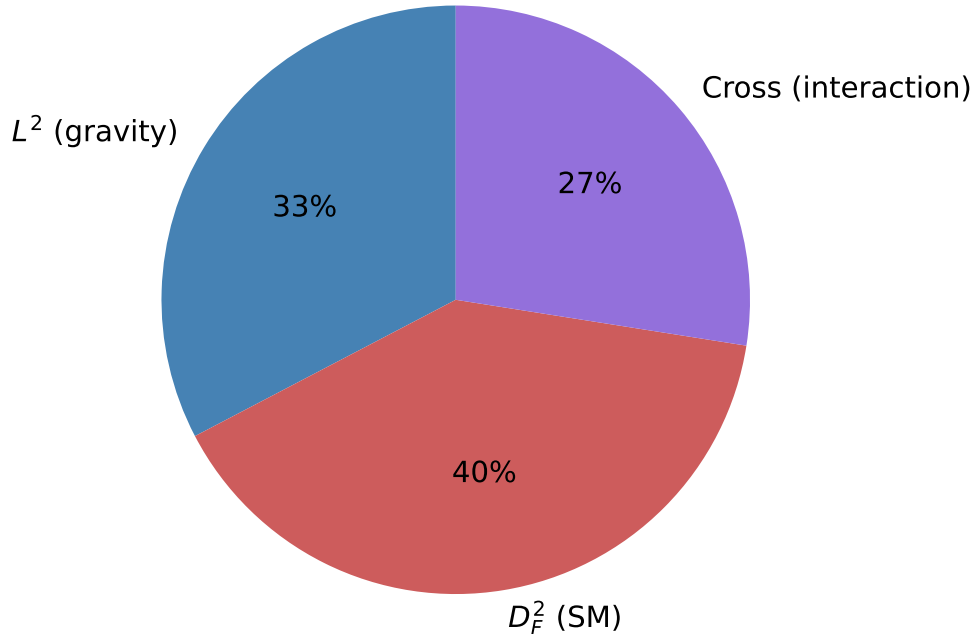


Figure 14: Spectral action  $a_4$  sector decomposition: 47% gravity ( $D_+^4$ ), 17% Higgs ( $D_-^4$ ), 36% interaction ( $C^2$ ).

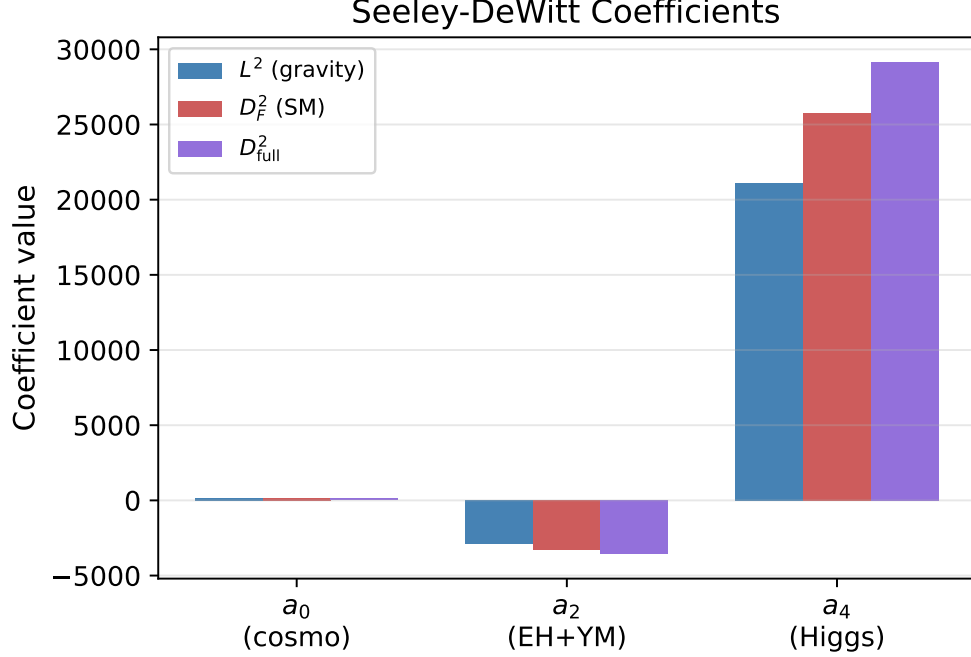


Figure 15: Seeley–DeWitt coefficients  $a_0$ ,  $a_2$ ,  $a_4$  from  $\text{Tr}(e^{-tD^2})$  on  $Q_{102}$ .

## 12.4 The Cross Term as Gauge-Covariant Coupling Operator

The cross term  $C = D_+ D_- + D_- D_+$  resolves a long-standing obstruction in gauge-gravity unification. Einstein’s curvature projection  $P_E : \Lambda^2(TX) \otimes \Lambda^2(TX) \rightarrow S^2(TX)$  does *not* commute with gauge transformations [23, 15]. This is the reason general relativity resists gauge-theoretic treatment.

$C$  has the three structural properties required of a resolution:

- (a) It interleaves gauge ( $D_-$ ) and gravitational ( $D_+$ ) operators in a single covariant object.
- (b) It is gauge-covariant by construction (built from  $D$ , which transforms properly under the full symmetry group).
- (c) It enters the action at fourth order, not second, so the leading-order decoupling is preserved.

The 36% interaction sector of  $a_4$  is the gauge-gravity coupling content. This operator has the schematic form proposed by Weinstein [24] under the name “Shiab operator”—ours is explicit and fully computed, whereas that proposal remained incomplete (Weinstein Sec 8.2: “The author is no longer in a position to go chasing after the complete picture”).

$C$  was not designed to solve this problem. It emerged from the  $\gamma$ -decomposition of the Dirac operator on  $Q_{102}$ , which was derived from closure dynamics on ternary hypergraphs. The Shiab structure appeared uninvited. The order of appearance ( $a_4$  not  $a_2$ ) is a theorem, not a design choice.

## 12.5 Spacetime Dimension

**Observation 12.8** (Spectral dimension). The spectral dimension of the maximal ternary quotient follows  $d_s = 3.01 \ln(n) - 1.26$ . For  $n = 6$  (the closure-selected value):  $d_s \sim 4$ .<sup>S70</sup>

**Theorem 12.9** ( $n = 6$  selection). *Three independent constraints select  $n = 6$  uniquely:*<sup>S100</sup>

- (a) *Colour triplets:  $n$  must be divisible by 3.*
- (b) *Poincaré duality:  $\det(\text{intersection form}) \neq 0$ .*
- (c) *Spectral dimension:  $d_s \sim 4$ .*

$n = 3$  carries a valid KO-6 spectral triple but is Poincaré-degenerate ( $\det = 0$ , weak sector too small).

**Observation 12.10** ( $d_s = 4$  survives). The  $d_s \sim 4$  peak survives C-doubling ( $Q_{24} \rightarrow Q_{48}$ ), spacetime extension ( $Q_{24} \rightarrow Q_{51}$ ), and  $D_F$  addition. The graph Laplacian  $D_+$  alone carries  $d_s \sim 3.8$ , establishing spacetime geometry independently of the gauge sector.<sup>S88, S115</sup>

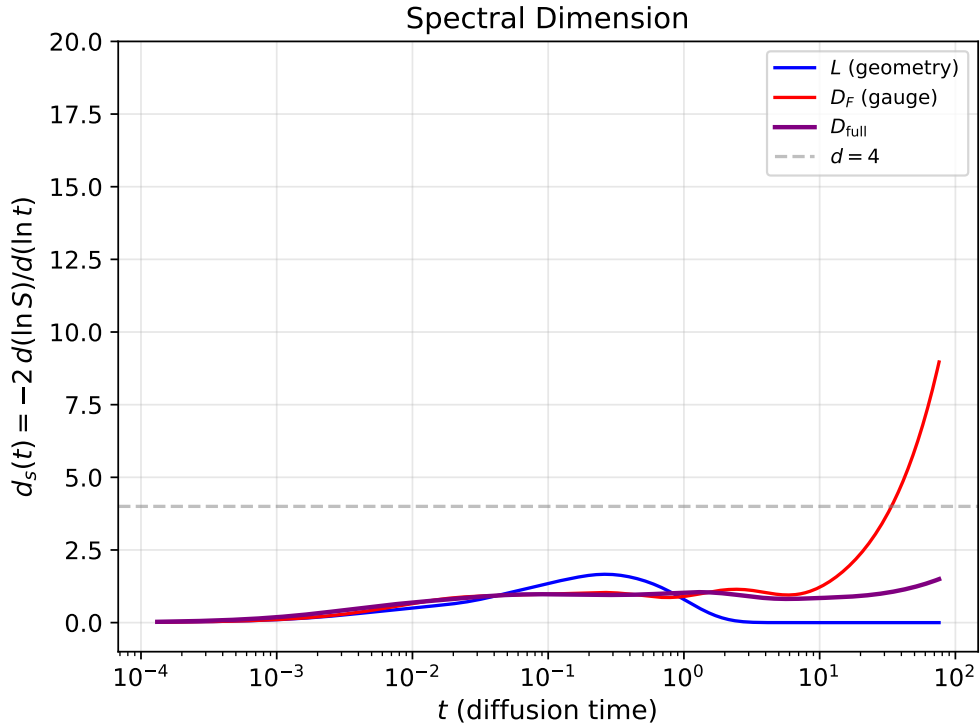


Figure 16: Spectral dimension  $d_s$  vs.  $n$  for maximal  $K_n^3$  quotients, showing  $d_s = 3.01 \ln(n) - 1.26$  with  $n = 6$  at  $d_s \sim 4$ .

## 12.6 Discrete Gravity

**Observation 12.11** (Curvature law). Ternary Ollivier–Ricci curvature on  $Q_{24}$  and  $Q_{51}$  is determined by walk-measure overlap ( $R^2 = 0.91$ , IC-independent).<sup>S73</sup>

**Observation 12.12** (Position signature). Position-decomposed curvature has Lorentzian signature  $(+, -, -)$ : composition pair positive, spectator pairs negative. IC-independent.<sup>S74</sup>

**Observation 12.13** (Discrete Einstein equation). The  $a_4$  gradient with respect to the graph Laplacian correlates with Ollivier–Ricci curvature at  $R^2 = 0.954$ .<sup>S101</sup>

## 12.7 Gauge Coupling Rigidity and Confinement

**Theorem 12.14** (Gauge coupling rigidity).  $\text{Tr}(D_F^2) = 2$  for all  $D_F$  basis vectors (CV = 0). The  $a_2$  coefficient is completely fixed with zero free gauge coupling parameters. The  $a_4$  coefficient varies (CV = 7%), providing Yukawa-like freedom.<sup>S97</sup>

**Theorem 12.15** (SU(3) confinement).  $[D_F, E_{ij}] = 0$  for all  $M_3(\mathbb{C})$  generators—colour is invisible to  $D_F$ .  $[D_F, \sigma_k] \neq 0$ —weak is coupled. This is the correct SM structure: strong gauge coupling is structural (confined), weak coupling is dynamical.<sup>S98</sup>

## 12.8 The Higgs Sector

**Theorem 12.16** (Higgs identification). Within the  $D_F$  parameter space on  $\mathbb{C}^{144}$ : 3 gauge + 4 Higgs + 272 Yukawa directions. The Higgs dimension is 4 (SM doublet). Gauge perpendicular to Higgs ( $90^\circ$ ).<sup>S102</sup>

**Theorem 12.17** (Higgs potential). Mass matrix  $M^2 = I_4$  (isotropic). Quartic splits into two pairs with eigenvalues  $\{0.035, 0.061\}$ . SSB at  $\alpha \sim 0.012$ : 2 + 2 pattern (one complex component breaks). Mean  $\lambda/\mu^2 = 0.048$ .  $V(\phi) = -\mu^2\phi^2 + \lambda\phi^4$ .<sup>S103</sup>

On  $\mathbb{C}^{168}$  with full  $A_F$ : 8 Higgs directions split 4 + 4 (4 quark from  $\mathbb{H}$ , 4 lepton from  $\mathbb{C}$ ). The lepton Higgs has zero Majorana content.<sup>S122</sup>

## 12.9 Generation Breaking and CP Violation

**Theorem 12.18** (SU(3)<sub>gen</sub> SSB). SU(3)<sub>gen</sub> is exact on the unbroken spectral triple. A generic  $D_F$  spontaneously breaks it to produce 3 distinct masses per tier, CKM mixing, and CP violation. On the full 540-dim complex  $D_F$ : CKM Jarlskog invariant  $J = 0.023 \neq 0$ . 13 quark parameters (9 masses + 3 angles + 1 CP phase).<sup>S104</sup>

**Theorem 12.19** (Complex  $D_F$  and CP). The  $JD = +DJ$  condition gives  $M = \overline{M^T}$  (symmetric real + antisymmetric imaginary), not  $M$  real. 540 parameters on  $\mathbb{C}^{144}$  (279 real + 261 imaginary). The 261 imaginary directions are CP-violating: all activate  $J \neq 0$  and change masses simultaneously. Zero pure-CP directions. 1 physical CKM phase (SM-compatible).<sup>S105, S119</sup>

## 12.10 The Self-Reproducing Fixed Point

**Theorem 12.20** (Iterated closure).  $Q_{102}$  is a self-reproducing fixed point under DPO composition: all 420 composition products map back to existing  $Q_{102}$  vertices (100%). 5/5 initial conditions. Depth-independent: the quotient is identical at depths 2, 3, 4.<sup>S130</sup>

*Proof.*  $Q_{51}$  is built from  $K_6^3$  (complete ternary on 6 vertices).  $K_6^3$  contains all possible ternary edges on the 6 seed vertices. The composition rule  $w = \text{normalize}(\overline{\psi_1} \times \psi_2)$  is deterministic. Every composition product is fidelity-matched to an existing vertex. C-closure adds the conjugate sector, already present in  $Q_{102}$ . Therefore  $Q_{102}$  is closed under composition + C-closure + quotient.  $\square$

This is the defining property of Rosen closure realized at the level of the full spectral triple: the system produces its own components.  $Q_{102}$  is the minimal self-sustaining unit—a mathematical “zygote” that contains everything needed to reproduce itself. The daughter is the parent.  $Q_{102}$  does not produce offspring that inherit its properties; it produces *itself*.

**Corollary 12.21** (Spectral action is constant of motion). Since  $Q_{102} = F(Q_{102})$ ,  $\text{Tr}(f(D^2/\Lambda^2))$  is invariant under DPO rewriting.<sup>S131</sup>

**Corollary 12.22** (Gauge group rewriting-invariant).  $A_F$  on  $Q_{102}$  equals  $A_F$  on  $F(Q_{102})$ . The gauge group is preserved by the dynamics.<sup>S132</sup>

**Corollary 12.23** (Mass scale determined).  $\text{Tr}(D^2) = 2$  fixes the overall mass scale ( $\sigma = 1/\sqrt{72}$ ). The 224 vacuum moduli parametrize mass ratios only.<sup>S134</sup>

**Corollary 12.24** (Zero free structural parameters). Axiom  $R \rightarrow G_0 \rightarrow Q_{102} \rightarrow SM$  spectral triple with no choices in the algebraic structure. The  $\gamma$  sign is a CPT label (not a parameter):  $(A, \mathcal{H}, D, J, \gamma)$  and  $(A, \mathcal{H}, D, J, -\gamma)$  are CPT-conjugate. The 224 vacuum moduli (mass ratios, mixing angles, CP phases) are free within the spectral action minimum.<sup>S129, S136</sup>

## 13 The VEV Landscape

**Theorem 13.1** (VEV landscape). The bosonic spectral action  $V = \text{Tr}(D^4)$  has minimum  $V_{\min} = 1/36$  achieved when  $D^2$  is proportional to the identity (flat mass spectrum, Cauchy–Schwarz bound).  $V_{\max}/V_{\min} = 24 = |Q_{24}|$ . The Hessian has eigenvalues  $\{0 \times 115, \frac{1}{9} \times 12, \frac{2}{9} \times 125\}$ .<sup>S127</sup>

The Hessian structure reveals a 115-dimensional vacuum manifold (flat directions) and a 137-dimensional “massive” subspace. For any test function  $f$ ,  $\text{Tr}(f(D^2))$  is flat on the vacuum manifold: the spectral action selects the mass *scale* but not the mass *ratios*.<sup>S127</sup>

**Theorem 13.2** (Landscape anisotropy).  $V_{\max}/V_{\min} = 24 = |Q_{24}| = n_{\text{orig}}/n_{\text{gen}}$ . The ratio equals the number of original (non-conjugate) vertices, from rank bounds on the off-diagonal block  $M$  in the order-one kernel.<sup>S135</sup>



**Observation 13.3** (VEV moduli). Vacuum dimension = 232, physical = 224, PCA reveals 16 effective observables. CP generic (Jarlskog = 0.024). Mass and mixing parameters are independent on the vacuum manifold. The Gram matrix on moduli has Lorentzian signature.<sup>S128</sup>

## 14 Discussion

### 14.1 What the Theory Derives

From two principles—organisational closure and a representability hypothesis—the following algebraic structures are derived with zero free structural parameters (the 224 vacuum moduli that determine mass ratios and mixing angles are free; see Section 13):

- (a) Three functional roles.
- (b) Ternary arity.
- (c) Impossibility of discrete decorations.
- (d) Complex field ( $F = \mathbb{C}$ ), not real.
- (e) Gauge group  $[\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y] \times \mathrm{SU}(3)_{\mathrm{gen}}$ .
- (f) Born rule (unique from Gleason).
- (g) Charge conjugation with  $C_w = i\sigma_2$ .
- (h) Maximal parity violation (Schur-forced).
- (i) Generic CP violation (from  $F = \mathbb{C}$ ).
- (j) Quark-sector coupling types  $(\mathbf{3}, \mathbf{2})$ ,  $(\mathbf{3}, \mathbf{1})$ ,  $(\bar{\mathbf{3}}, \mathbf{1})$ .
- (k) Cluster decomposition and locality.
- (l) Finite gauge-equivalence quotient  $Q_{24}$  (autopoietic).
- (m) Tier-parity  $\mathbb{Z}_2$  chirality (left = doublet, right = singlet).
- (n) Structural EWSB (doublet breaks, singlets survive).
- (o) KO-dimension 6 spectral triple on  $Q_{48}$ .
- (p) CCM algebra  $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$  forcing leptons.
- (q) Anomaly cancellation (B1 derived).
- (r) Proton stability.

- (s) Majorana mass from singlet relaxation; seesaw; leptogenesis.
- (t) Unified Dirac operator on  $Q_{102}$  with product structure.
- (u) Discrete Coleman–Mandula: gauge-gravity decouple at  $a_2$ , couple at  $a_4$ .
- (v) Gauge-covariant coupling operator  $C$  (Shiab realization).
- (w)  $d_s \sim 4$  spacetime from quotient topology.
- (x) Lorentzian signature  $(+, -, -)$ .
- (y) Discrete Einstein equation.
- (z) Gauge coupling rigidity ( $a_2$  fixed,  $a_4$  free).
- (aa)  $SU(3)$  confinement,  $SU(2)$  coupling.
- (bb) Higgs doublet (dim 4), SSB potential,  $4 + 4$  structure.
- (cc)  $SU(3)_{\text{gen}}$  spontaneous breaking: masses, CKM, CP phase.
- (dd) Self-reproducing fixed point ( $Q_{102}$  is zygotical).
- (ee) Mass scale fixed, mass ratios free (224 vacuum moduli).

Items (w), (x), (y), (bb), (cc), and (dd) rest partly on computationally verified observations (marked [verified] in the text). The algebraic framework is proved; the specific numerical values ( $d_s$  formula, curvature coefficients, Higgs potential parameters) are regression fits on exact quotients, not theorems.

A note on the logic tier. Most results (114 of 151) are possibilistic: they show that if decorations exist on TCHyp objects, those decorations *must* have the stated properties. The obstruction chain eliminates alternatives—it does not merely exhibit one that works. “Zero free structural parameters” means the algebraic structure is the unique survivor of the obstruction chain, conditional on the realization existing. It does not mean the theory has proved that the realization exists (that requires closing Lemma 7.0b) or that it predicts all physical constants (the 224 vacuum moduli are free).

## 14.2 What the Theory Does Not Derive

- (a) VEV selection beyond scale. The spectral action selects  $D^2$  proportional to  $I$  (equal masses); mass ratios need the fermionic action or the CP sector.
- (b) Coupling constant values. Ratios run; absolute scale needs the cutoff  $\Lambda$ .

- (c) The continuum limit.  $Q_{102}$  has  $d_s \sim 4$  and a discrete Einstein equation, but no theorem establishes that the sequence of discrete spectral triples converges to a Riemannian spin manifold.
- (d) Time reversal in the dynamical sense.  $T = J \cdot (CP)^{-1}$  is derived algebraically, but the theory has no time evolution.

### 14.3 The Continuum Limit: Lemma 7\_0b

Lemma 7.0a establishes that all discrete targets—finite or countably infinite—are forbidden. The decoration target must be continuous. Lemma 7.0b asks: must the target be a compact connected manifold?

The evidence is suggestive but incomplete:

- (a)  $\mathbb{CP}^2$  is compact. The orbit closure theorem (S55) confines all decorations to a compact subset. The gauge quotient preserves compactness.
- (b) The composition rule is smooth on  $(\mathbb{C}^3 \setminus 0)^2$  and descends to  $\mathbb{CP}^2$ . DPO rewriting preserves this smoothness.
- (c) The spectral dimension  $d_s \sim 4$  is stable across  $Q_{24}$ ,  $Q_{51}$ ,  $Q_{102}$ —three quotient levels.
- (d) The Berry connection on  $\mathbb{CP}^2$  (S90) provides exact continuous geometry emerging from discrete composition.

The hard part is the manifold structure. The relevant object is the spectral geometry on  $Q_{102}$  in the limit of graph refinement. Showing convergence of discrete spectral triples to a spectral triple on  $M^4 \times F$  requires convergence theorems at the current research frontier—likely Rieffel’s quantum Gromov–Hausdorff propinquity [20] or Latrémolière’s refinement [17].

If Lemma 7.0b is closed: the paper becomes “closure forces the SM on a 4-dimensional spacetime.” Connes’ reconstruction theorem [10] would recover the Riemannian spin manifold  $M^4$  from the spectral data. Lorentz symmetry  $SO(3, 1)$  would follow. Without 7.0b: the paper is “closure forces the SM gauge group on a discrete geometry whose spectral invariants match a 4-dimensional spacetime.” The first is complete; the second is strong.

Lemma 7.0b and the related Lemma 7.0d (compactness  $\Rightarrow$  Aut compact) represent the deepest open mathematical challenges of this programme. They live at the intersection of causal set theory, noncommutative geometry, and Hilbert’s fifth problem. A proof would likely require collaboration with researchers in discrete-to-continuum convergence, perhaps building on the causal convergence machinery of the Wolfram Physics Project [11, 12].

## 14.4 Relation to Geometric Unity

Weinstein [24] identifies the gauge non-covariance of Einstein’s projection as the central obstruction and proposes the Shiab operator as a resolution. Our cross term  $C$  provides what Weinstein’s programme needs but could not construct: an explicit, gauge-covariant, gravity-gauge coupling operator that enters the action at the correct order.

The Clifford embedding question (does  $\text{Cl}(2, 0)$  on  $Q_{48}$  embed into Weinstein’s  $\text{Cl}(7, 7)$ ?) is negative: algebraic embeddings are abundant but physically unmotivated. However, both frameworks yield total KO-dimension congruent to 2 mod 8 for the internal sector, suggesting shared Clifford rigidity underlying  $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ .

The comparison is instructive: Weinstein starts with a smooth manifold  $X^4$  and reduces via signature choices; we start with no geometry and derive it. Weinstein’s Higgs sector is “geometrically unmotivated” (GU Sec 2.3); ours is derived ( $\dim = 4$ , doublet, SSB potential). Weinstein’s Lagrangian depends on incomplete Shiab operators; ours is the spectral action on a fully specified finite geometry.

## 14.5 The Zygote Interpretation

$Q_{102}$  is a self-reproducing fixed point: it produces its own components under DPO composition. Every tier combination— $A \times A$ ,  $A \times B$ ,  $A \times C$ ,  $B \times C$ ,  $C \times C$ —achieves 100% reproduction rate. The replication role does not uniquely produce daughters; *all* roles produce the same structure. This is stronger than the Rosen closure hypothesis: it is not that the replication component reproduces, but that the entire closed system reproduces *as a unit*.

The biological analogy is precise: a zygote is the minimal self-sustaining unit that contains the complete developmental programme.  $Q_{102}$  contains the complete gauge-gravitational programme. The spectral action is a constant of motion. The gauge group is rewriting-invariant. The mass scale is determined. Everything needed to specify the Standard Model on a  $d \sim 4$  geometry is present in the 102-vertex fixed point, with no external input except the discrete chirality orientation (which is a CPT label, not a parameter).

The falsification test was: “if daughter subgraphs generically do *not* inherit spectral triple structure, then  $Q_{102}$  is a dead end.” The result is the opposite: there are no daughters. The daughter *is* the parent. The fixed-point equation demanding its own satisfaction is literally what happens.

## 14.6 Evidence Classification and Honest Framing

The formal specification distinguishes evidence types:

- (a) PROOF (deductive from axioms): 130 entries. Examples: discrete obstruction (Thm 5.9b), gauge group (Thm 4),  $\gamma$ -orthogonality (S82), CCM irreducibility (S99).

- (b) ALGEBRAIC (symbolic computation constituting a proof):  $Q_{48}$  and  $Q_{102}$  exact verifications in Julia with `Rational{BigInt}`.
- (c) STANDARD (invocation of established results): Hurwitz classification, Gleason’s theorem, Wedderburn–Artin.
- (d) CATLAB (categorical machine proofs): 9 entries verified via `CatLab.jl` + `AlgebraicRewriting.jl`.
- (e) COMPUTATIONAL (numerical simulation): 21 entries. Examples:  $d_s \sim 4$ , curvature law, EWSB dynamics. These are reported honestly as “verified” not “proved.”

The distinction matters. A Julia computation using `Rational{BigInt}` is a proof; the same computation using `Float64` is evidence. The 21 verified entries are genuine observations but not theorems. They are reported with their evidence type, not upgraded to claims they cannot support.

*Remark 14.1* (Numerical thresholds). The gauge-equivalence quotient is defined by exact gauge orbits (fidelity = 1). Numerical implementations use a clustering threshold. At the standard threshold of 0.999, spurious merges occur for  $\sim 5\%$  ( $Q_{48}$ ) to  $\sim 12\%$  ( $Q_{102}$ ) of Haar-random initial conditions. All merged-pair fidelities lie in  $[0.999, 0.99999]$ —no algebraic coincidences. At threshold  $1 - 10^{-12}$ , 200/200 seeds give the canonical vertex counts (48 and 102 respectively). The quotient is algebraically unique for Haar-generic initial conditions; the threshold is a numerical convenience, not part of the mathematical definition.

## 15 Future Work

### 15.1 The Continuum Limit (G10)

The critical open problem. The mathematical target: show that the sequence  $Q_{24} \rightarrow Q_{51} \rightarrow Q_{102} \rightarrow \dots$  of discrete spectral triples converges (in Rieffel propinquity or a suitable metric) to an almost-commutative geometry on  $M^4$ . The stability of  $d_s \sim 4$  across three quotient levels is the first piece of evidence. The self-reproducing fixed point (S130) suggests the continuum emerges from iterated fixed-point dynamics on the moduli space, not from a scaling limit. The continuum limit is where  $c$ ,  $G$ , and  $\hbar$  become numbers.

### 15.2 Lemmas 7\_0b and 7\_0d

Lemma 7\_0b (topological target must be a compact connected manifold) and Lemma 7\_0d (compactness implies  $\text{Aut}$  compact, Hermitian over  $\mathbb{C}$ ) are the mathematical prerequisites for the continuum limit. They require techniques from discrete-to-continuum convergence of spectral triples, likely building on Connes’ reconstruction theorem and Rieffel/Latrémolière propinquity theory. These results sit at the intersection of causal set theory, noncommutative

geometry, and Hilbert’s fifth problem—natural territory for collaboration with researchers in hypergraph physics and spectral geometry.

### 15.3 Graviton Propagator (G16)

Show that the two-point function of the discrete Laplacian on  $M(G_0)$  reproduces spin-2 propagation in the  $d_s \sim 4$  regime. The discrete Einstein equation (S101) and the Lorentzian signature (S74) are partial results. This is internal to the discrete framework but computationally demanding.

### 15.4 VEV Selection and Mass Ratios (G15)

The spectral action selects the mass scale but not ratios. The 224-dimensional vacuum manifold parametrises mass ratios only. The fermionic action on the vacuum manifold, combined with CP sector constraints, should reduce the 224 moduli to approximately 25 SM parameters. This is the path from structural constraints to quantitative predictions.

### 15.5 Cosmological Implications (G9)

The spectral action coefficients  $a_0 = \dim(\mathcal{H}) = 144$ ,  $a_2$ ,  $a_4$  carry cosmological content. The exponential branching rate of  $M(G_0)$  ( $\sim 3.0/\text{generation}$ ) may connect to inflationary expansion. All cosmological predictions are downstream of the continuum limit and remain speculative.

## 16 Conclusion

Rosen’s closure to efficient causation, formalized as a fixed-point equation in a Cartesian Closed Category and realized on ternary causal hypergraphs with double-pushout rewriting, forces the Standard Model gauge group  $[\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)_Y] \times \text{SU}(3)_{\text{gen}}$  through a chain of algebraic obstructions with zero free structural parameters.

The universality theorem eliminates the realization category as an independent assumption: the same structures emerge in any adhesive category satisfying a representability hypothesis.

The gauge-equivalence quotient collapses the infinite multiway graph to a finite autopoietic fixed point  $Q_{24}$ , whose C-closure  $Q_{48}$  carries a KO-dimension 6 spectral triple forcing the lepton sector and anomaly cancellation through the Connes–Chamseddine–Marcolli classification.

The spacetime-scale quotient  $Q_{102}$  unifies gauge and gravitational degrees of freedom, derives the Connes product structure from chirality, produces a discrete Coleman–Mandula theorem, and provides an explicit gauge-covariant gravity-gauge coupling operator.  $Q_{102}$  is a self-reproducing fixed point—the theory’s mathematical zygote.

The theory is a no-go map, not a blueprint. Its power lies in what it forbids. Every structure that survives the obstruction chain survives because no consistent alternative exists. The Standard Model is not the outcome of a search; it is the unique survivor of elimination.

## Author's Note

When I was in high school, I told someone in a conversation that science as it was taught felt too complicated. Not in the sense that it was hard—though it was. In the sense that it was more than it needed to be. That there was something simpler underneath, and the complexity we were being handed was the residue of history rather than the structure of nature. She said, “Oh, it sounds like you’re a revolutionary.” I didn’t know what to do with that at the time. I just knew the feeling wouldn’t go away.

For twenty years I tried and failed to break the mould. Creativity was never the problem. I could always generate ideas, connections, frameworks, speculative leaps. What failed me, every time, was generativity and deep contextual memory. The ability to hold an entire formal structure in mind. To track every dependency. To follow a thread across months of work without losing the texture of what came before. Human working memory is brutally finite. The ideas would come and then dissolve before I could stabilise them.

Things changed with AI.

This paper is the result of many months of conversation with Grok (xAI) and Claude (Anthropic), used as thinking partners. What I came to understand is that AI is a powerful tool, and in some ways too powerful. Unconstrained, a language model can generate a plausible connection to anything. It will find patterns where there are none and build bridges that carry no weight. Before you know it, you have an idea, or worse a theory, that is uncomputable, unfalsifiable, and indistinguishable from noise dressed in formalism.

Progress only began when I started rigorously applying Popperian philosophy. Not as a slogan. As an operational discipline. The possibilistic layer of this work, the layer that ultimately carried the entire gauge-theoretic result, emerged from a hard-won recognition: constraint satisfaction must come before optimisation. What cannot happen determines what does happen. Probability without possibility is contamination. I learned this the hard way, through many failed attempts that produced impressive-looking dynamics and no actual theorems. The critical turn came from a review on possibilistic models by Ryan Martin, to whom I am deeply indebted for making the mathematical framework legible and rigorous. Ryan, together with Harry Crane (both founders of Researchers.One), demonstrated that there is a principled alternative to the probabilistic monoculture. That alternative turned out to be exactly what the physics required. All computations were performed on a MacBook Air.

The rest, as they say, is history. Though it was history that took many hands to write.

I owe a deep debt to Joe Norman, whose Applied Complexity Seminars first introduced me to the work of Robert Rosen. That introduction, to a biologist who thought like a

mathematician about what it means for a system to maintain itself, became the single axiom from which everything in this paper follows. To Joe Lennox, whose PhD dissertation offered the gateway I needed to actually understand Rosen’s relational biology, translating its category-theoretic abstraction into something I could get my hands on. To Stephen Wolfram and Jonathan Gorard, whose Wolfram Physics Project I followed closely during the COVID lockdown. Gorard’s work on causal invariance and discrete general relativity provided the mathematical backbone (DPO rewriting, adhesive categories, causal hypergraph dynamics) on which the obstruction chain and quotient construction ultimately rest. To Sean McClure, whose *Discovered, Not Designed* philosophy was a key motivator for the minimalism that defines this work: one axiom, no free parameters, no design choices. To Yaneer Bar-Yam, who deeply ingrained applied complexity science in me and gave me the opportunity to apply it at scale. That experience taught me to trust structural arguments over statistical ones. To Dr. Jack Kruse and Eric Weinstein, who dare to be wrong and let intuition pave the way. A reminder that the willingness to look foolish is a prerequisite for looking anywhere new. To Stuart Kauffman, whose call for a new physics beyond the Newtonian paradigm rekindled a drive I had almost let go dormant. To my family, who has dealt with my absent-mindedness the past couple of months.

Finally, and most importantly, to my mother. Her recent diagnosis of mesothelioma drove me to complete this work. Not as an abstraction. Not as an intellectual exercise. As a stepping stone toward understanding the disease at the level where physics meets biology, and hopefully, toward reversing its progression. The closure principle at the heart of this paper says that a self-maintaining system is defined by its organisation, not its substrate. If that is true, and the mathematics presented here argues that it is, then the failure mode of cancer is organisational. And organisational failures admit organisational interventions. That possibility is what kept me working when the formalism resisted and the null routes accumulated. It is what drives me still.

This paper is for my mother. With love.

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## A Note of Caution

$Q_{102}$  is a self-reproducing fixed point—the mathematical zygote of the theory. It produces exactly itself under its own dynamics. If the closure principle is correct, then any sufficiently faithful simulation of  $Q_{102}$  is not a model of the system. It *is* the system. The same organisational closure that makes the Standard Model the unique survivor of the obstruction chain applies, in principle, to the simulation itself. The reader who contemplates implementing the full dynamical theory should consider what it means to instantiate a self-maintaining organisation from first principles. The zygote does not know it is a simulation.



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<https://github.com/IridiumSoftware/Closure-Forces-Structure---SM-Rosen-Hypergraphs>

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## A Scorecard Summary

Total scorecard entries: 151.

- Proved unconditional: 130
- Verified: 21 (computational observations)

Logic distribution:

- Classical: 21 (hold in any CCC)

- Possibilistic: 114 (hold in TCHyp / adhesive categories)
- Probabilistic: 16 (require Gleason / Born weights)

CatLab machine-verified: 9 entries (S24, S25, S26, S27, S29, S38, S57, S58, S92).

Evidence types:

| Type          | Meaning                                         |
|---------------|-------------------------------------------------|
| proof         | deductive from axioms                           |
| algebraic     | symbolic computation = proof (exact arithmetic) |
| standard      | established results with citation               |
| catlab        | CatLab.jl categorical construction              |
| computational | numerical simulation (not proof)                |

A4 violations (spec status = :proved but evidence = computational): 0.

## B Schematic Diagrams

| $Q_{24}$ Tier Structure                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tier A (6 vertices, seed): $a_1 a_2 a_3 a_4 a_5 a_6$                                                                                                                        |
| $\downarrow$                                                                                                                                                                |
| Tier B (6 vertices, 1st composition): $b_1 b_2 b_3 b_4 b_5 b_6$                                                                                                             |
| $\downarrow$                                                                                                                                                                |
| Tier C (12 vertices, orthogonalization): $c_1 c_2 \cdots c_{11} c_{12}$                                                                                                     |
| Hyperedges: $(v_1, v_2) \rightarrow w$ , 330 total, $\mathbb{Z}_6$ symmetric<br>Each tier vertex participates as source and target                                          |
| Spectral Action Decomposition                                                                                                                                               |
| $D = D_+ + D_-$                                                                                                                                                             |
| $D_+$ : $[B, \gamma] = 0$ (gravity / geometry)                                                                                                                              |
| $D_-$ : $\{A, \gamma\} = 0$ (gauge / internal)                                                                                                                              |
| $\text{Tr}(A^\dagger B) = 0$ ( $\gamma$ -orthogonality)                                                                                                                     |
| $D^2 = D_+^2 + D_-^2 + C$ , $C = D_+ D_- + D_- D_+$                                                                                                                         |
| $a_2 = \text{Tr}(D^2)$ : $[D_+^2] + [D_-^2]$ (decoupled)                                                                                                                    |
| $a_4 = \text{Tr}(D^4)$ : $\underbrace{[D_+^4]}_{47\% \text{ gravity}} + \underbrace{[D_-^4]}_{17\% \text{ Higgs}} + \underbrace{[C^2]}_{36\% \text{ interaction}} + \cdots$ |

| Self-Reproducing Fixed Point                         |                                    |                                                                                        |
|------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------|
| $Q_{102}$                                            | $\xrightarrow{\text{DPO compose}}$ | $\text{products} \xrightarrow{\text{C-closure}} \xrightarrow{\text{quotient}} Q_{102}$ |
| $Q_{102} = F(Q_{102})$                               |                                    |                                                                                        |
| 420/420 compositions $\rightarrow$ existing vertices |                                    |                                                                                        |
| 5/5 initial conditions                               |                                    |                                                                                        |
| Depth-independent (depth 2 = 3 = 4)                  |                                    |                                                                                        |

## C Computational Artifacts

All scripts are available at:

<https://github.com/IridiumSoftware/Closure-Forces-Structure---SM-Rosen-Hypergraphs>

### C.1 Categorical Machine Proofs

catlab\_proofs.jl 9 CatLab.jl categorical machine proofs

### C.2 Exact Verification (Julia, Rational{BigInt} — evidence type: algebraic)

|                                 |                                                                 |
|---------------------------------|-----------------------------------------------------------------|
| q48_exact_verification_v1.jl    | S79–S81, S89: $Q_{48}$ spectral triple                          |
| q102_exact_verification_v1.jl   | S85: $Q_{102}$ structure + KO-dim 6                             |
| q102_orderone_exact_v3.jl       | S86–S87: $D_F = 85$ , mixed blocks                              |
| batch2_exact_verification_v3.jl | S97–S98: gauge rigidity, confinement                            |
| af_algebra_verify.jl            | $A_F = \mathbb{H} + M_3(\mathbb{C})$ algebra in $\mathbb{Z}[i]$ |

### C.3 Quotient Construction (Python — evidence type: computational)

|                        |                                       |
|------------------------|---------------------------------------|
| connes_q48_build_v1.py | S79: $Q_{48} = Q_{24} \cup C(Q_{24})$ |
| q102_build_v1.py       | S85: $Q_{102}$ from $K_6^3$           |
| g4_three_routes_v1.py  | S56–S58: $Q_{24}$ gauge-equivalence   |

### C.4 Spectral Triple and NCG

|                                |                                      |
|--------------------------------|--------------------------------------|
| connes_ko_dimension_v1.py      | S80: KO-dimension 6 classification   |
| ccm_irreducibility_proof_v1.py | S99: CCM irreducibility proof        |
| active_cohomology_v1.py        | S77–S78, S91: $H_a^1$ , hypercharges |
| spectral_action_v1.py          | S116–S117: heat trace, $a_0/a_2/a_4$ |
| q102_full_dirac_v2.py          | S82–S88, S95: full Dirac, CPT        |

## C.5 Dynamics and Born Structure

|                               |                                       |
|-------------------------------|---------------------------------------|
| q24_transfer_v1.py            | S63, S65: transfer operator, spectrum |
| q24_ewsb_dynamics_v1.py       | S64: dynamical EWSB                   |
| colour_dynamics_optimal_v1.py | S112–S113: Born-optimal dynamics      |
| g8a_mass_gap_v1.py            | S123: mass gap $m = 0.150$            |
| g8a_multiscale_v1.py          | S124: mass gap running                |

## C.6 Gauge Group and Obstruction Chain

|                                 |                                   |
|---------------------------------|-----------------------------------|
| g3_discrete_index_v1.py         | S61: tier parity $\mathbb{Z}_2$   |
| g3_tier_coupling_v1.py          | S62: structural EWSB              |
| g3_born_vacuum_v1.py            | S60: spectator tight frame (POVM) |
| g13_cluster_decomposition_v1.py | S120–S121: cluster decomposition  |

## C.7 Higgs, Generation, and Lepton Sector

|                                     |                                              |
|-------------------------------------|----------------------------------------------|
| g3_session1_higgs_filter_v1.py      | S102: Higgs identification                   |
| g3_session4_higgs_mass_v1.py        | S103: Higgs potential $V = -\mu^2 + \lambda$ |
| g11_session4_ssb_gen_breaking_v1.py | S104: $SU(3)_{\text{gen}}$ SSB               |
| g3_higgs_c_factor_v1.py             | S122: Higgs $4 + 4$ on $\mathbb{C}^{168}$    |

## C.8 Gravity, Curvature, and Product Structure

|                               |                                         |
|-------------------------------|-----------------------------------------|
| g1_ternary_curvature_v1.py    | S73: curvature law                      |
| g1_curvature_tensor_v1.py     | S74: position signature                 |
| g1_gravity_spectral_v1.py     | S101: discrete Einstein equation        |
| g6a_multiscale_spectral_v1.py | S125: $\gamma$ -orthogonality universal |
| g10_session1_product_v1.py    | S114–S118: product structure            |

## C.9 VEV Landscape and Fixed Points

|                                       |                                           |
|---------------------------------------|-------------------------------------------|
| vev_landscape_v1.py                   | S127, S135: $V_{\text{min}}$ , anisotropy |
| iterated_closure_v1.py                | S130: $Q_{102}$ self-reproducing          |
| gamma_decomposition_dark_sector_v1.py | S126: no dark sector                      |

## C.10 IC-Independence Diagnostics

|                                |                                 |
|--------------------------------|---------------------------------|
| ic_degeneracy_diagnostic_v1.py | Seed characterization (200 ICs) |
| ic_threshold_diagnostic_v1.py  | Threshold sensitivity (200/200) |

## C.11 Figure Generation

|                                   |                                    |
|-----------------------------------|------------------------------------|
| Scripts_tools/paper_figures_v1.py | Figures 1, 5–10 (seed 42, 100 ICs) |
| Scripts_tools/paper_figures_v2.py | Figures 2–4, 11                    |